

Thimza: Stateless Threshold Signatures at Optimal Bandwidth

Abstract

Although the Raccoon (CRYPTO’24) has made a promising impact among lattice-based signatures, all of its variants hide partial signatures behind one-time additive masks. This mechanism brings up three issues. Initially, a repeated mask leaks the key share scaled by a Lagrange coefficient, thus a signer must remember every session identifier it has ever used; removing this unbounded state left as an open problem by del Pino et al. (EUROCRYPT’24). Secondly, the security proofs, by default, assume authenticated channels realized by message authentication codes and, in the two-round case, a signer set fixed before the message is known. The only attempt to remove this assumption inflates signatures from 11 to 30–50 kilobytes, and closing that parameter gap flagged as an open problem by Zhu-Tessaro (CRYPTO’25). Thirdly, at a threshold of $t = 1024$ signers, communication per signer ranges from 27 to 613 kilobytes, and 71 kilobytes even for the most recent scheme built for medium committees, which is inefficient.

In this work, we address we address all three issues. We start by characterizing statefulness precisely, interactive Fiat–Shamir threshold signature can not be stateless, and one ephemeral self-erasing token per session is necessary and sufficient. Then, we prove lemma on view partitions to show that masks derived from a signer’s own transcript view remain simulatable under adversarially inconsistent views, which removes authenticated channels and the preprocessing aware of the signer set at no cost in parameters. Next, build upon our results, we present Thimza¹, the most time-efficient lattice-based threshold signature and the most space-efficient among the interchangeable schemes. Thimza sends 16.3 KiB per signer at $t = 1024$, a factor 1.6 below the previous best and a factor 2.5 below the best prior scheme based on the same assumptions, and whose 12.8 KiB signatures are accepted by the unmodified verifier of the underlying single signer scheme. Evaluations show that Thimza signs between 1.6 and 20.2 times faster at $t = 1024$, and 6 times faster than the most recent medium scale design; our Thimza is even faster than Ringtail (S&P’25) and Hermine (NIST MPTC’26). In addition to formal security proofs, we also provide machine-checked security guarantees for Thimza in EasyCrypt toolset.

Keywords

Threshold Signatures, One-Time Masking, Module Short Integer Solution, Lattices, Post-Quantum Cryptography

1 Introduction

A (t, n) -threshold signature distributes a signing key among n parties so that any t of them can jointly produce a signature that an ordinary verifier accepts, while $t - 1$ of them learn nothing useful [2]. The migration of this primitive to post-quantum assumptions is a stated priority of the ongoing call for multi-party threshold schemes [3], and the lattice-based line of work initiated by Threshold Raccoon [4] has produced, in three years, a family of schemes [5–10] that are practical for a threshold as large as $t = 1024$.

All members of that family share one mechanism. The signing key s , a short vector over a cyclotomic ring, is shared with Shamir’s scheme [11], so that party i holds a share s_i and contributes $c \cdot \lambda_{T,i} \cdot s_i$ to the response, where c is the Fiat–Shamir challenge and $\lambda_{T,i}$ is the Lagrange coefficient of party i for the signer set T . Because $\lambda_{T,i}$ is an essentially uniform ring element, this contribution is not short and cannot be flooded by the ephemeral randomness; it must therefore be hidden by a one-time additive mask m_i , chosen so that the masks of the signer set sum to zero [4]. Most of the disadvantages about these schemes follows from this single mechanism.

1.1 Issues and gaps

Problem 1: unbounded signer state. The masks must never repeat. In Threshold Raccoon [4] the mask of pair (i, j) is $\text{PRF}(\text{seed}_{i,j}, \text{sid})$ for a session identifier sid , so a signer has to store the set of all identifiers it has ever answered and refuse a repetition. That set grows without bound, and a rollback of the store, for instance the restoration of a virtual machine snapshot, is fatal. Developers of Threshold Raccoon mention that their scheme requires to maintain the unique identifier for each signing session so as not to reuse the same masks and they leave a stateless signature scheme as an important open problem [4]. No subsequent work has settled whether statelessness is achievable, and the scheme with two rounds [5, 6, 8] inherit a session state whose necessity has never been analyzed.

Problem 2: authenticated channels, preprocessing aware of the signer set, and the parameter gap. The masks of two honest signers cancel only if the two signers hold the same view of the transcript. Threshold Raccoon [4] therefore makes every signer authenticate its view with a message authentication code, and Ringtail [6] assumes authenticated channels together with an offline oracle that already knows the signer set T ; developers of Ringtail state that removing these constraints while retaining a proof from standard assumptions is a direction for future work [6]. The restrictions were removed by reproving the scheme of [5] under an algebraic one-more module-SIS problem, which does reduce to MLWE and MSIS, but only in a large-parameter regime, where signature sizes grow from about 11 KiB to 30–50 KiB [8, 10]. The developers observe that closing the gap between the cryptanalysis-driven parameter choices and those derived from their security reduction remains an open problem [9]. The question is therefore whether the strong network model can be obtained at the parameters of the weak one.

Problem 3: bandwidth. For $t = 1024$ at the 128-bit level, communication per signer is 40.8 KiB for Threshold Raccoon [4], 276 KiB for the scheme with two rounds of [5], 613 KiB for Ringtail [6], and 26.9 KiB for TalonG [8], the last of which is not verifiable by the plain Raccoon verifier and additionally assumes NTRU. No scheme in the family is simultaneously the smallest in communication, the smallest in signature size, and compatible with the verifier of the single signer scheme.

1.2 Our contributions

A solution for the statelessness question (Section 3). We formalize the state used for partial signing of a threshold signature and give a dichotomy. On the negative side we show that no threshold signature in the *canonical linear-response* class, which contains every

¹The name *Thimza* derives from the abbreviation “Th” for **T**hreshold and the Turkish word *imza*, meaning “signature” [1], so Thimza: Threshold Signature.

Fiat–Shamir lattice construction we are aware of as well as FROST [12], admits a stateless algorithm for partial signing, since an adversary that replays the last round under two different views recovers the key share with two online queries (Theorem 3.3). On the positive side we show that the unbounded identifier registry of [4] is not needed. If the one-time mask is derived from the signer’s own view rather than from a session identifier, then replaying an identical view returns an identical response, so a signer only has to consume each ephemeral token once and may erase it immediately (Lemma 3.5 and Corollary 3.6). This is the exact minimum permitted by Theorem 3.3, and it resolves the problem of [4] which was left open.

Unauthenticated channels and preprocessing free of the signer set at unchanged parameters (Section 4). We define Threshold Signature Unforgeability with Channel Agnostic and Coalition Agnostic oracles (TS-UF-CA), an unforgeability game in which the adversary delivers arbitrary and mutually inconsistent messages of the first round to the honest signers and in which no oracle knows the signer set before the message is fixed. Our main technical tool is the view-partition lemma (Lemma 4.2), namely that under view-derived masking the honest responses decompose along the partition of the honest signers induced by their views, every class that does not contain all honest signers is masked by an unmatched pseudorandom term, and only the all-honest class carries information. Consequently, a simulator that can answer sessions in which the views agree can answer arbitrary ones, and message authentication codes become unnecessary. Combining this with a commitment free of the signer set we obtain a direct reduction to MLWE and SelfTargetMSIS at the parameters of the weak model (Theorem 4.3), which closes the gap identified in [9] for the profile with three rounds. We complement this with a matching lower bound, namely that any non-interactive pairwise masking mechanism needs $t - 1$ full-entropy pseudorandom vectors per signer per session (Theorem 4.4), and a single seed per unordered pair attains it, which reduces the cost of all prior schemes.

Shamir Secret Sharing $\rightarrow$ [31] (Shamir, How to Share a Secret, 1979), Raccoon Signature $\rightarrow$ [16] (Raccoon, CRYPTO 2024; the threshold version is built on Threshold Raccoon [15]), SHAKE256 (Hash/PRF) $\rightarrow$ no dedicated reference is cited in the bibliography despite being specified as the instantiation in Appendix N, and the Threshold-signing protocol basis $\rightarrow$ [15] (Threshold Raccoon, EUROCRYPT 2024).

The Thimza scheme (Section 5–Section 7). We present Thimza, a (t, n) threshold signature over module lattices whose signatures are Raccoon signatures, so a threshold signature is accepted by the unmodified verifier of the single signer scheme. Thimza, built upon building blocks Shamir Secret Sharing [11] + Raccoon Signature [13] + SHAKE256 (Hash/PRF) [14] + Threshold-signing protocol [4], combines antisymmetric one-time masking, view-derived mask derivation, a first round free of the signer set, and a new transmission of the leading bits of the commitment rule under which each signer publishes only the top $\lceil \log_2 q \rceil - v_c$ bits of its commitment. At the 128 bit level and $t = 1024$ each signer sends 16 736 bytes, against 57 184 for Threshold Raccoon [4], 278 400 for the scheme of [5], 629 248 for Ringtail [6] and 27 546 for TalonG [8], while the signature stays at 12.8 KiB and the verification key at 3.8 KiB; the recent scheme Hermine [15], which targets committees of at most 64 signers, sends 73 216 bytes. We implement Thimza and comparable schemes in a single framework and measure a signing time per

signer that is 1.56 times below Threshold Raccoon [4], 2.29 times below TalonG [8], 9.1 times below the scheme of [5] and 20.2 times below Ringtail [6] at $t = 1024$, and 6.0 times below Hermine [15] at $t = 64$.

1.3 Technical overview

The starting point is the observation that the three problems have a common root, namely the coupling of the mask to a session identifier. Once the mask is a function of the signer’s own view, a repeated view yields a repeated response and therefore no new information, which removes the identifier registry; distinct views yield masks whose pairwise terms no longer cancel, which removes the need to authenticate views. What remains is to show that the second effect does not break the simulation, and this is exactly the content of the view-partition lemma. The lemma is proved by exhibiting, for every partition class that is not the all-honest class, a pseudorandom term that occurs with a single sign in the sum of that class, which makes the class sum uniform.

The bandwidth improvement is independent of the above and comes from a budget analysis. In a three-round Fiat–Shamir threshold signature the message of the second round $\mathbf{w}_i = \mathbf{A}\mathbf{r}_i + \mathbf{e}'_i$ is only ever used inside the rounded aggregate $\lfloor \sum_j \mathbf{w}_j \rfloor_{v_w}$, and the rounding error introduced by truncating each $\mathbf{w}_i$ is absorbed by the signature hint. Because the aggregate noise already has standard deviation $\sigma_w \sqrt{t}$, truncating v_c least significant bits perturbs the hint by $2^{v_c} \sqrt{t}/12$, and the two are balanced at $v_c = 36$ for the parameters of [4], which shrinks the message of the second round by a factor 3.8. In the scheme with two rounds the same message is multiplied by a wide Gaussian aggregation vector before rounding, which is why no comparable truncation is available there; we analyze it in Section 6.

1.4 Related work

As Table 1 shows, before this work no scheme was at once smallest in communication, competitive in signature size and interchangeable, but Thimza achieves that point.

Generic constructions from threshold fully homomorphic encryption [16, 17] are round-optimal but require homomorphic evaluation of the signing circuit. Direct constructions in the Fiat–Shamir with aborts paradigm [18–20] were first obtained for the n -out-of- n case. Threshold Raccoon [4] gave the first efficient (t, n) scheme without heavy tools, at three rounds; [5] and [6] reduced this to two rounds with a preprocessable first round, the former under an algebraic one-more assumption and the latter under MLWE with a lattice trapdoor embedded in the matrix of the first round; [9] reproved both from MLWE and MSIS. [7] avoids pairwise masks by using a linear secret sharing with small coefficients, at a large cost in share size. [8] uses an NTRU trapdoor semi-commitment to obtain two rounds with low bandwidth, at the price of a first round that depends on the message, an extra assumption, and a signature that carries the commitment opening and is therefore not accepted by the plain Raccoon verifier. Adaptive security is addressed in [10, 21] and identifiable abort in [22]. Sedghighadikolaie-YavuzA [23] have done a broad survey of this area in their work.

Closest to our design point is Hermine [15], which removes the one time masks altogether by sharing the signing key with a Vandermonde scheme whose shares are themselves short, so that a partial signature can be verified on its own. This buys two properties that no other scheme in the family has, namely identifiable abort

without extra interaction and proactive refresh of the shares, and it keeps a first round that can be preprocessed. The price is paid on three points. The shares are short rather than uniform, so their number grows with the committee and the scheme is instantiated for at most 64 signers, where our own measurement of its key material gives 255 KiB per signer at $(n, t) = (16, 8)$. The first round must carry $\text{rep} = 13$ commitment columns, which is 53 248 bytes, and the response covers $\ell + k$ ring elements instead of ℓ . And the proof goes through the algebraic one more module SIS problem rather than directly through MLWE and SelfTargetMSIS. Thimza and Hermine therefore sit at two ends of one trade: Hermine spends bandwidth and scale to buy accountability, and Thimza spends accountability to buy bandwidth, speed and scale.

2 Preliminaries

2.1 Notation

We write λ for the security parameter, $[n] = \{1, \dots, n\}$, and $x \leftarrow D$ for sampling x from a distribution D . Let φ be a power of two, $\mathcal{R} = \mathbb{Z}[X]/(X^\varphi + 1)$ the cyclotomic ring of degree φ , q a prime with $q \equiv 1 \pmod{2\varphi}$, and $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ the quotient ring, which supports a number theoretic transform. Elements of $\mathcal{R}$ are identified with their coefficient vectors, and $\|x\|_p$ denotes the ℓ_p norm of that vector; the notation extends to vectors and matrices over $\mathcal{R}$ by concatenation. We write D_σ for the discrete Gaussian of width parameter σ over $\mathbb{Z}$, extended coefficientwise, and $\eta_\epsilon(\Lambda)$ for the smoothing parameter of a lattice Λ [27]. For $v \in \mathbb{N}$ and $x \in \mathbb{Z}_q$ with unsigned representative in $[0, q)$, the rounding operator $\lfloor x \rfloor_v = \lfloor x/2^v \rfloor \bmod q_v$ with $q_v = \lfloor q/2^v \rfloor$ drops the v least significant bits, and extends coefficientwise. The challenge set is $C = \{c \in \mathcal{R} : \|c\|_\infty = 1, \|c\|_1 = \omega\}$, the set of ternary polynomials of Hamming weight ω .

2.2 Lattice assumptions

We use three assumptions, all standard in this line of work.

Definition 2.1 (Module learning with errors [28]). For a matrix dimension pair (k, ℓ) , a noise width σ and a number of samples k , the MLWE $_{q,k,\ell,\sigma}$ problem asks to distinguish $(A, As + e)$ from (A, u) , where $A \leftarrow \mathcal{R}_q^{k \times \ell}$ is uniform, the secret $s \leftarrow D_\sigma^\ell$ and the error $e \leftarrow D_\sigma^k$ are short, and $u \leftarrow \mathcal{R}_q^k$ is uniform.

Definition 2.2 (Self-target module SIS [29]). For a norm bound β and a hash function $H : \mathcal{R}_q^k \times \{0, 1\}^* \rightarrow C$ modelled as a quantum-accessible random oracle, the SelfTargetMSIS $_{q,k,\ell,C,\beta}$ problem asks, given a uniform $A \leftarrow \mathcal{R}_q^{k \times \ell}$ and superposition access to H , to output a message msg and a vector $y = (z, h, c)$ with $\|y\|_2 \leq \beta$ and $H([A \mid I_k] \cdot (z, h), \text{msg}) = c$.

Definition 2.3 (Hint module learning with errors [30]). For a number of hints Q , challenge distribution $\mathcal{L}$ and widths σ_1, σ_2 , the Hint-MLWE $_{q,k,\ell,Q,\sigma_1,\sigma_2,\mathcal{L}}$ problem is MLWE where the distinguisher additionally receives Q hints $(c_j, c_j s + r_j, c_j e + e'_j)$ with $c_j \leftarrow \mathcal{L}$ and $r_j, e'_j \leftarrow D_{\sigma_2}$.

Kim et al. [31] give a reduction from MLWE with width σ_0 to Hint-MLWE whenever $\frac{1}{\sigma_0} \geq 2(\frac{1}{\sigma_1} + \frac{1}{\sigma_2})$ and $\sigma_2 \geq \eta_\epsilon(\mathbb{Z}^\varphi)\sqrt{B}$ with $B \geq \sum_j \|c_j\|_1^2$; the reduction is a sequence of statistically close distributions and therefore carries over to a quantum distinguisher. We take SelfTargetMSIS as a hardness assumption in its own right, as [32] does for the same parameters, since the forking lemma

of [33] rewinds the adversary and has no counterpart for superposition queries, and MSIS is implied by MLWE for the parameters we use.

2.3 Shamir over $\mathcal{R}_q$ and Raccoon signature

For distinct evaluation points $\alpha_1, \dots, \alpha_n \in \mathbb{Z}_q^\times$ whose pairwise differences are units, sharing a secret $s \in \mathcal{R}_q^\ell$ with threshold t means sampling $p_1, \dots, p_{t-1} \leftarrow \mathcal{R}_q^\ell$ uniformly and setting $s_i = s + \sum_{u=1}^{t-1} p_u \alpha_i^u$. For $|T| = t$ the values of Lagrange coefficients $\lambda_{T,i} = \prod_{j \in T \setminus \{i\}} \alpha_j(\alpha_j - \alpha_i)^{-1}$ satisfy $\sum_{i \in T} \lambda_{T,i} s_i = s$, and any $t-1$ shares are uniform and independent of s .

The Raccoon signature [4, 13] is Lyubashevsky's Fiat-Shamir construction [34] without rejection, made short by noise flooding. The verification key is $\mathbf{t} = \lfloor A\mathbf{s} + \mathbf{e} \rfloor_{v_t}$ for a uniform matrix $A \in \mathcal{R}_q^{k \times \ell}$, a short secret $s \leftarrow D_{\sigma_1}^\ell$ and a short error $e \leftarrow D_{\sigma_1}^k$. A signature is a triple (c, z, h) consisting of a challenge, a response and a hint, and verification accepts if

$$c = H(\text{vk}, \lfloor A\mathbf{z} - 2^{v_t} \mathbf{t} \rfloor_{v_w} + \mathbf{h}, \text{msg}) \wedge \|(z, 2^{v_w} \mathbf{h})\|_2 \leq B, \quad (1)$$

where v_t and v_w are the numbers of bits dropped from the key and from the commitment, and B is the verification bound.

2.4 Threshold signatures

Definition 2.4 (Threshold signature scheme). A (t, n) -threshold signature is a tuple $\Sigma = (\text{KeyGen}, \text{Sign}_1, \dots, \text{Sign}_\rho, \text{Combine}, \text{Verify})$ where $\text{KeyGen}(1^\lambda, t, n)$ outputs a verification key vk and signing keys $(\text{sk}_i)_{i \in [n]}$, each Sign_u is the algorithm run by a signer in round u of a ρ -round protocol, Combine assembles the round messages into a signature, and Verify is the verification algorithm. The scheme is *functionally interchangeable* with a scheme with a single signer Σ' if $\text{Verify} = \Sigma'.\text{Verify}$ and every honestly produced signature lies in the support of $\Sigma'.\text{Sign}$ for the same key.

Unforgeability is the standard [35] game with static corruption as the adversary picks a corruption set C with $|C| < t$, receives $(\text{vk}, (\text{sk}_i)_{i \in C})$, interacts with the honest signers through per-round oracles, and wins if it outputs a valid signature on a message never submitted to the signing oracles. We write $\text{Adv}_{\Sigma, \mathcal{A}}^{\text{ts-uf}}(\lambda)$ for its advantage. Figure 1 states the game for a protocol with three rounds; the two boxed restrictions are the ones we remove in Section 4.

As Figure 1 presents, the adversary first fixes a corruption set C below the threshold and receives the keys of those parties, the set $\mathcal{H}$ holds the honest indices, $\mathcal{M}$ records every message an honest signer ever answered, the three oracles are the three protocol rounds, and eid is the handle of one open session, which is the token of Definition 3.4. The boxed argument T forces a signer to know the signer set before it commits, and the boxed check forbids the adversary from speaking on behalf of an honest signer, which a deployment realizes with message authentication codes; both are removed in Section 4.

3 The state of a threshold signer

Here, we address the problem of [4]. We first isolate the class of schemes in which the question aims, then show that statelessness is unachievable in that class, and finally show that the unbounded identifier registry is nonetheless avoidable.

Definition 3.1 (Canonical linear-response scheme). A ρ -round threshold signature is *canonical linear-response* if there are efficiently computable functions Cmt , Chl and Rsp such that, for every

Table 1: Lattice-based threshold signatures at the 128-bit level. Sizes in KiB. “|tr|/t” is communication per signer of one signing execution, “on” the part of it that cannot be preprocessed, “rd/on-rd” number of rounds and of rounds that depend on the message, and interchangeability “interch.” whether a signature is accepted by unmodified verifier of single signer scheme.

Scheme	t	rd/on-rd	tr /t	on	vk	σ	interch.	Assumptions
Bendlin et al. [24] (ACNS’13)	small	≥ 4	$> 10^3$	all	—	—	✗	SIS (generic MPC)
Boneh et al. [16] (CRYPTO’18)	any	1	$> 10^3$	all	—	—	✗	LWE (threshold FHE)
Agrawal et al. [17] (ICALP’22)	any	1	$> 10^3$	all	—	—	✗	LWE (threshold FHE)
Damgård et al. [18] (PKC’21)	$t = n$	3/3	25.0	all	7.0	12.0	✗	MLWE, MSIS
Chen [19] (CRYPTO’23)	$t = n$	2/1	300.0	30.9	10.5	30.9	✗	MLWE, MSIS
Threshold Raccoon [4] (EUROCRYPT’24)	1024	3/3	40.8	all	3.9	12.7	✓	MLWE, MSIS
Espitau et al. [5] (CRYPTO’24)	1024	2/1	276.1	14.1	5.5	10.8	✓	AOM-MLWE, MSIS
Gur et al. [25] (PQCrypto’24)	5	2/2	3072.0	all	13.0	46.0	✗	RLWE, RSIS
C-Apirom et al. [7] (ASIACRYPT’24)	5	2/1	1526.0	≈ 260	47.0	258.0	✗	MSIS
Ringtail [6] (IEEE S&P’25)	1024	2/1	612.8	10.5	4.5	13.4	✓	MLWE, MSIS
del Pino, Niot [26] (PKC’25)	8	3/3	7.5	all	2.6	2.7	✗	MLWE, MSIS
TalonG [8] (EUROCRYPT’26)	1024	2/2	26.9	all	2.0	17.7	✗	RLWE, RSIS, NTRU
Hermine [15] (NIST MPCT’26)	64	2/1	71.5	19.5	2.8	11.0	✓	AOM-MISIS
Thimza (Ours)	1024	3/3	16.3	all	3.8	12.8	✓	MLWE, MSIS

Experiment $\text{Exp}_{\Sigma, \mathcal{A}}^{\text{ts-uf}}(1^\lambda)$

- (1) $C \leftarrow \mathcal{A}(1^\lambda)$; abort if $|C| \geq t$ or $C \not\subseteq [n]$.
- (2) $(vk, (sk_i)_{i \in C}) \leftarrow \text{KeyGen}(1^\lambda, t, n)$; set $\mathcal{H} := [n] \setminus C$, $\mathcal{M} := \emptyset$.
- (3) $(\text{msg}^*, \sigma^*) \leftarrow \mathcal{A}^{O_1, O_2, O_3, H}(vk, (sk_i)_{i \in C})$.
- (4) Return 1 iff $\text{Verify}(vk, \text{msg}^*, \sigma^*) = 1$ and $\text{msg}^* \notin \mathcal{M}$.

Oracles (for $i \in \mathcal{H}$)

- $O_1(i, \text{msg}, \boxed{T})$: sample a fresh token, store $(\text{eid}, i, \text{st}_i)$, return the message of the first round and eid.
- $O_2(\text{eid}, i)$: return the message of the second round; mark eid opened.
- $O_3(\text{eid}, i, T, (\text{cmt}_j, \tilde{w}_j)_{j \in T})$: abort if eid is closed;

abort if some $j \in T \cap \mathcal{H}$ did not produce $(\text{cmt}_j, \tilde{w}_j)$;

 set $\mathcal{M} := \mathcal{M} \cup \{\text{msg}\}$, close eid, return z_i .

Figure 1: Unforgeability with static corruption. The boxed input of O_1 makes the first round coalition-aware and the boxed check models authenticated channels; Definition 4.1 removes both.

signer i , the message of the first round is $\text{Cmt}(sk_i; \rho_i)$ for fresh randomness ρ_i , the challenge used in the last round is $c = \text{Chl}(\text{vw}_i)$ where vw_i is the transcript that signer i has received, and the message of the last round is

$$z_i = c \cdot \gamma_i(\text{vw}_i) \cdot s_i + f_i(\rho_i, \text{vw}_i) + m_i(\text{vw}_i), \quad (2)$$

where $\gamma_i(\text{vw}_i) \in \mathcal{R}_q$ is a public reconstruction coefficient, f_i is independent of s_i , and m_i is a masking term computable from sk_i and vw_i .

Threshold Raccoon, the schemes of [5–8], Thimza, and FROST [12] in the group setting all satisfy Definition 3.1, with $\gamma_i = \lambda_{T,i}$ and f_i the ephemeral randomness of signer i , possibly transformed by a public aggregation vector.

Definition 3.2 (Stateless partial signing). A canonical linear-response scheme has *stateless partial signing* if every signer’s algorithm of the last round is a deterministic function of (sk_i, vw_i) alone, so that a signer can answer any number of requests in the last round without consulting or updating any persistent one.

The next statement addresses the impossibility of statelessness for interactive threshold signatures.

THEOREM 3.3. *Let Σ be a canonical linear-response threshold signature with $\rho \geq 2$ rounds in which the randomness of the first round ρ_i is sampled before the view that determines the challenge is complete. If Σ has stateless partial signing, then there is an adversary that corrupts $t - 1$ parties, makes one query of the first round and two queries of the last round to a single honest signer h , and recovers $\gamma_h s_h$ up to a public invertible factor with probability at least $1 - \phi/q - 2^{-\lambda}$. In particular Σ is not unforgeable whenever $t \leq n$.*

Proof sketch. The adversary obtains the message of the first round of h , then sends two views vw, vw' that agree on the signer set but differ in one corrupt commitment, so that $\gamma_h(\text{vw}) = \gamma_h(\text{vw}') = \gamma$ while $c = \text{Chl}(\text{vw}) \neq c' = \text{Chl}(\text{vw}')$. Statelessness forces h to answer both with the same ρ_h , so subtracting the two responses of (2) cancels $f_h(\rho_h, \cdot)$ up to a term the adversary can compute, and leaves $(c - c')\gamma s_h$ plus the difference of the masks. Since h is the only honest signer in T , the masking terms of all other members of T are computable by the adversary and the zero-sum property determines m_h , so the difference of masks is known. The element $c - c'$ is invertible in $\mathcal{R}_q$ except with probability at most ϕ/q , which yields γs_h . The full argument, including the case $f_h(\rho_h, \text{vw}) \neq f_h(\rho_h, \text{vw}')$, is in Appendix A. $\square$

Theorem 3.3 says that some per-session state is unavoidable. The following definition captures the weakest form of state that is compatible with it.

Definition 3.4 (Session-bounded state). A threshold signature has *session-bounded state* if a signer’s persistent store consists only of tokens created in the first round of an open session, each token is consumed by at most one execution of the last round and erased immediately afterwards, and the store is empty whenever no session is open.

Threshold Raccoon does not have session-bounded state, since its store contains every session identifier ever used and never shrinks, because an adversary may open a fresh session that reuses an old identifier and thereby a mask. The next lemma is what removes that requirement. We call a masking term *view-derived* if $m_i(\text{vw}_i) =$

$\sum_{j \in T \setminus \{i\}} \varepsilon_{ij} \text{PRF}(\text{sd}_{\{i,j\}}, \text{vw}_i)$ with $\varepsilon_{ij} = -\varepsilon_{ji} \in \{\pm 1\}$ and one seed $\text{sd}_{\{i,j\}}$ per unordered pair.

LEMMA 3.5. *Let Σ be a canonical linear-response scheme with view-derived masking, and let a signer h hold a fixed token ρ_h . If h is invoked twice in the last round on views vw and vw' with $\text{vw} = \text{vw}'$, then the two responses are equal. Consequently the second invocation is perfectly simulatable from the first, and grants the adversary no information beyond it.*

PROOF. By Definition 3.1 the response is $z_h = c\gamma_h(\text{vw})s_h + f_h(\rho_h, \text{vw}) + \mathbf{m}_h(\text{vw})$. All three summands are deterministic functions of $(\text{sk}_h, \rho_h, \text{vw})$: the challenge is $c = \text{Chl}(\text{vw})$ and the coefficient $\gamma_h(\text{vw})$ are functions of vw ; the term $f_h(\rho_h, \text{vw})$ is a function of the token and the view; and, since the masking is view-derived, $\mathbf{m}_h(\text{vw}) = \sum_j \varepsilon_{hj} \text{PRF}(\text{sd}_{\{h,j\}}, \text{vw})$ is a function of sk_h and vw because the seeds are part of sk_h and a pseudorandom function is deterministic. Since $\text{vw} = \text{vw}'$ and ρ_h is unchanged, all three summands take the same values in the two invocations, hence the responses are equal. A value that equals one already delivered is simulatable by copying it, so the adversary's view after the second invocation is a deterministic function of its view after the first. $\square$

The next statement addresses the equivalence between a session-bounded store and an unbounded identifier registry.

COROLLARY 3.6. *A canonical linear-response scheme with view-derived masking is unforgeable with session-bounded state if and only if it is unforgeable with a store that additionally records every session identifier. Moreover, session-bounded state is minimal, in the sense that no smaller store suffices by Theorem 3.3.*

PROOF. One direction is immediate because a larger store can only reject more queries. For the other direction, consider an adversary $\mathcal{A}$ against the scheme with session-bounded state. Every query of the last round of $\mathcal{A}$ either consumes an unconsumed token, in which case the scheme that records identifiers answers identically, or attempts to reuse a consumed token. A reuse with the same view is answered with the same value by Lemma 3.5 and can be simulated by $\mathcal{A}$ itself from its transcript; a reuse with a different view is rejected in both schemes, in the session-bounded scheme because the token was erased. Hence the two games are perfectly indistinguishable and the advantages coincide. Minimality follows because Theorem 3.3 shows that a store which does not prevent a second consumption of the same token with a different view is insecure, and a store which prevents it must retain at least one bit per open token. $\square$

3.1 Insufficiency of one identifier

It is instructive to see that the registry of [4] cannot be replaced by a smaller device without changing the mask derivation, because this explains why Lemma 3.5 needs the view and not merely a counter. Suppose a signer keeps only a monotone counter and refuses identifiers below it, so that each identifier is used once but the mask depends on the identifier alone. An adversary opens a session with identifier sid , lets the honest signer h complete it, and then opens a *new* session with the same sid but a fresh token; the counter does not detect this, since the identifier is being used for the first time by the honest signer in that new session only if the counter was reset, and if it was not reset the adversary simply picks sid from a namespace the counter does not order, for instance one supplied by a peer. The two sessions then use the same mask

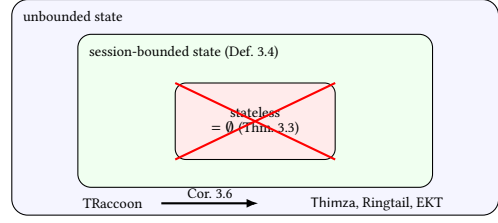


Figure 2: The three regimes of signer state and the results that separate them.

$\mathbf{m}_h(\text{sid})$ but different signer sets $T \neq T'$, hence different Lagrange coefficients. Subtracting the two responses removes the mask and leaves

$$z_h - z'_h = (c\lambda_{T,h} - c'\lambda_{T',h})s_h + \mathbf{r}_h - \mathbf{r}'_h, \quad (3)$$

and since $\lambda_{T,h}$ and $\lambda_{T',h}$ are essentially uniform ring elements chosen by the adversary through its choice of T , the coefficient of s_h is uniform and not short, so the flooding provided by $\mathbf{r}_h - \mathbf{r}'_h$, of width $\sigma_w\sqrt{2}$, is useless: $\ell\varphi$ repetitions give a full-rank linear system over $\mathcal{R}_q$ and recover s_h . This is why the registry must be a set and not a counter. With view-derived masking the two sessions have different views, so the masks differ and (3) never becomes available; the only replay that a signer must refuse is a second consumption of the same token, which by Lemma 3.5 it may do by simply erasing it.

As Figure 2 presents, the three regimes of signer state are nested. The outer region is the unbounded one, where the store grows with the number of executions and where Threshold Raccoon lies. The middle region is that of Definition 3.4, where the store holds one token per open session and returns to empty, and where Thimza and, through Corollary 3.6, the schemes with two rounds lie. The inner region is stateless, and Theorem 3.3 shows it to be empty for interactive schemes; only round optimal constructions from threshold fully homomorphic encryption reach it. The arrow carries the result that moves a scheme between regions.

4 Views, channels and the cost of masking

4.1 A stronger unforgeability game

The unforgeability games of [4, 6] restrict the adversary in two ways, since it must deliver to every honest signer the genuine messages of the first round of the other honest signers, which in practice is realised by message authentication codes, and in the two-round case the offline oracle already takes the signer set T as input. We remove both.

Definition 4.1 (Unforgeability with oracles agnostic to the channel and to the signer set). Let Σ be a (t, n) -threshold signature with three rounds. For a quantum adversary $\mathcal{A}$ with superposition access to the oracles $H \in \{H_0, H_1, H_c\}$, the experiment $\text{Exp}_{\Sigma, \mathcal{A}}^{\text{ts-uf-ca}}(1^\lambda)$ is the one of Figure 3, and

$$\text{Adv}_{\Sigma, \mathcal{A}}^{\text{ts-uf-ca}}(\lambda) := \Pr \left[\text{Exp}_{\Sigma, \mathcal{A}}^{\text{ts-uf-ca}}(1^\lambda) = 1 \right].$$

The scheme Σ is TS-UF-CA secure if $\text{Adv}_{\Sigma, \mathcal{A}}^{\text{ts-uf-ca}}$ is negligible in λ for every such $\mathcal{A}$ running in quantum polynomial time.

As Figure 3 presents, C is the statically corrupted set, $\mathcal{H}$ the honest indices, $\mathcal{M}$ the messages an honest signer ever answered, and St the set of session records, each of which carries the handle eid , the owner i , the message, the token st_i and a flag. The two differences

Experiment $\text{Exp}_{\Sigma, \mathcal{A}}^{\text{ts-uf-ca}}(1^\lambda)$	
(1)	$C \leftarrow \mathcal{A}(1^\lambda)$; abort if $C \not\subseteq [n]$ or $ C \geq t$.
(2)	$(\text{vk}, (\text{sk}_i)_{i \in [n]}) \leftarrow \text{KeyGen}(1^\lambda, t, n)$.
(3)	$\mathcal{H} := [n] \setminus C$; $\mathcal{M} := \emptyset$; $\text{St} := \emptyset$; $\text{ctr} := 0$.
(4)	$(\text{msg}^*, \sigma^*) \leftarrow \mathcal{A}^{O_1, O_2, O_3, H_0\rangle, H_1\rangle, H_C\rangle}(\text{vk}, (\text{sk}_i)_{i \in C})$.
(5)	return $\llbracket \text{Verify}(\text{vk}, \text{msg}^*, \sigma^*) = 1 \rrbracket \wedge \llbracket \text{msg}^* \notin \mathcal{M} \rrbracket$.
Oracle $O_1(i, \text{msg})$ <i>T is not an input</i>	
(1)	if $i \notin \mathcal{H}$ then return $\perp$.
(2)	$\text{ctr} := \text{ctr} + 1$; $\text{eid} := \text{ctr}$.
(3)	$(\text{cmt}_i, \text{st}_i) \leftarrow \text{Sign}_1(\text{sk}_i, \text{msg})$.
(4)	$\text{St} := \text{St} \cup \{(\text{eid}, i, \text{msg}, \text{st}_i, \text{open})\}$.
(5)	return $(\text{eid}, \text{cmt}_i)$.
Oracle $O_2(\text{eid})$	
(1)	if $(\text{eid}, i, \text{msg}, \text{st}_i, \text{open}) \notin \text{St}$ then return $\perp$.
(2)	return $\text{Sign}_2(\text{st}_i)$.
Oracle $O_3(\text{eid}, T, (\text{cmt}_j, \bar{\text{w}}_j)_{j \in T})$ <i>no consistency check</i>	
(1)	if $(\text{eid}, i, \text{msg}, \text{st}_i, \text{open}) \notin \text{St}$ then return $\perp$.
(2)	if $i \notin T$ or $T \not\subseteq [n]$ or $ T \neq t$ then return $\perp$.
(3)	$\mathbf{z}_i \leftarrow \text{Sign}_3(\text{sk}_i, \text{st}_i, T, (\text{cmt}_j, \bar{\text{w}}_j)_{j \in T}, \text{msg})$.
(4)	$\text{St} := (\text{St} \setminus \{(\text{eid}, i, \text{msg}, \text{st}_i, \text{open})\}) \cup \{(\text{eid}, \text{closed})\}$; erase st_i .
(5)	$\mathcal{M} := \mathcal{M} \cup \{\text{msg}\}$.
(6)	return $\mathbf{z}_i$.

Figure 3: The game TS-UF-CA of Definition 4.1.

Table 2: Security models of the Raccoon family.

Scheme	auth. ch.	set at rd. 1	state	Assumption
TRaccoon [4]	required	required	$O(Q_s)$	MLWE, MSIS
Ringtail [6]	required	required	$O(1)$	MLWE, MSIS
EKT [5]	no	no	$O(1)$	AOM-MLWE
EKT via [9]	no	no	$O(1)$	MLWE, MSIS (3–5 $\times$)
Thimza	no	no	$O(1)$	MLWE, MSIS

from Figure 1 are the whole content of the definition. The oracle O_1 takes no signer set, so the message of the first round is produced before T exists, and O_3 performs no test relating $(\text{cmt}_j, \bar{\text{w}}_j)_{j \in T}$ to what the honest signers actually sent, so the adversary may supply forged, reordered or mutually inconsistent openings, and in particular may call O_3 on two honest signers of the same session with different vectors. Step 4 of O_3 closes the record before returning, which is exactly the requirement of Definition 3.4 that each token be consumed at most once, and step 1 of O_3 enforces it, so a replay of a closed handle returns $\perp$. Writing $|H\rangle$ for superposition access, the adversary queries the three hash functions on arbitrary quantum states, while the three signing oracles are classical because they are executed by honest parties.

Table 2 places the resulting model next to those of prior work. As Table 2 shows, the second column asks whether the proof needs channels an adversary cannot tamper with, which a deployment realizes with a message authentication code on every message of the first round; the third asks whether a signer must know the signer set before that round, which decides whether preprocessing can be shared across coalitions; the fourth is the size of the persistent store of Section 3, with Q_s the number of signatures ever produced; and the last names the assumption with the penalty it carries, where “3–5 $\times$ ” records that the reduction of [9] reaches the strong model from standard assumptions but with signatures of 30 to 50 KiB

in place of 11 KiB. Only the Thimza in last row has $O(1)$ and no penalty at once.

4.2 The view-partition lemma

Fix a session, let H denote the honest members of T and let $\sim$ be the equivalence on H that identifies two signers with the same view. Write $H/\sim = \{S_1, \dots, S_m\}$ for the induced partition into *view classes*. The following lemma addresses the behavior of view-derived masking under inconsistent views.

LEMMA 4.2. *Let the masking be view-derived with one seed per unordered pair and antisymmetric signs, and model PRF as a pseudorandom function. For every view class S_a with $S_a \neq H$, the sum $\sum_{i \in S_a} \mathbf{z}_i$ is, given the adversary’s entire view, computationally indistinguishable from uniform over $\mathcal{R}_q^\ell$. If $m = 1$, that is if all honest signers hold the same view, then $\sum_{i \in H} \mathbf{z}_i$ equals $c \sum_{i \in H} \lambda_{T,i} s_i + \sum_{i \in H} f_i$ plus a quantity computable by the adversary.*

PROOF. Consider first a class $S_a \subsetneq H$ and pick $h \in S_a$ and $h' \in H \setminus S_a$; such h' exists because $S_a \neq H$. The seed $\text{sd}_{\{h, h'\}}$ is shared by two honest signers, so it is unknown to the adversary. Signer h evaluates $\text{PRF}(\text{sd}_{\{h, h'\}}, \text{vw}_h)$ and signer h' evaluates $\text{PRF}(\text{sd}_{\{h, h'\}}, \text{vw}_{h'})$, and $\text{vw}_h \neq \text{vw}_{h'}$ because h and h' lie in different classes. Hence the value $v := \varepsilon_{hh'} \text{PRF}(\text{sd}_{\{h, h'\}}, \text{vw}_h)$ occurs exactly once in the multiset of all pseudorandom terms evaluated in the session, namely inside $\mathbf{m}_h$, and it occurs inside $\sum_{i \in S_a} \mathbf{z}_i$. Every other term of $\sum_{i \in S_a} \mathbf{z}_i$ is independent of v . By pseudorandomness of PRF under the key $\text{sd}_{\{h, h'\}}$, which the adversary never obtains, v is computationally indistinguishable from a uniform element of $\mathcal{R}_q^\ell$, and a sum in which a uniform independent summand occurs is itself uniform. This proves the first claim, with a distinguishing advantage bounded by Adv^{PRF} .

For the second claim, suppose $m = 1$ so that all honest signers hold the common view vw . Then

$$\sum_{i \in H} \mathbf{m}_i(\text{vw}) = \sum_{i \in H} \sum_{j \in T \setminus \{i\}} \varepsilon_{ij} \text{PRF}(\text{sd}_{\{i, j\}}, \text{vw}).$$

Split the inner sum according to whether $j \in H$ or $j \in T \setminus H$. Every unordered pair $\{i, j\} \subseteq H$ contributes the two terms $\varepsilon_{ij} \text{PRF}(\text{sd}_{\{i, j\}}, \text{vw})$ and $\varepsilon_{ji} \text{PRF}(\text{sd}_{\{i, j\}}, \text{vw})$, which cancel because $\varepsilon_{ij} = -\varepsilon_{ji}$ and the two evaluations use the same key and the same input vw . What remains is $\sum_{i \in H} \sum_{j \in T \setminus H} \varepsilon_{ij} \text{PRF}(\text{sd}_{\{i, j\}}, \text{vw})$, and every seed occurring there is shared with a corrupt signer and therefore known to the adversary. Adding the definition (2) of the responses gives the stated form. $\square$

Lemma 4.2 is the technical reason why authenticated channels can be dropped. A simulator that is able to produce the honest responses when all honest signers agree can produce them in general, because for every class other than the all-honest class it outputs uniform elements, and it uses its single simulation strategy only on the all-honest class, which is exactly the situation the proofs of [4, 6] already handle.

The next statement addresses signatures unforgeability in the strengthened model.

THEOREM 4.3. *Let Σ be a canonical linear-response scheme with three rounds with view-derived antisymmetric masking, a first round free of the signer set, and the Raccoon verification relation (1). If Σ is unforgeable in the game of Section 2 with authenticated channels and oracles aware of the signer set, then Σ is unforgeable in the game of Definition 4.1, and for every quantum adversary $\mathcal{A}$ making Q_s*

classical signing queries and Q_h quantum superposition hash queries there are adversaries $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ with

$$\begin{aligned} \text{Adv}_{\Sigma, \mathcal{A}}^{\text{ts-uf-ca}} &\leq \text{Adv}_{\mathcal{B}_1}^{\text{Hint-MLWE}} + \text{Adv}_{\mathcal{B}_2}^{\text{SelfTargetMSIS}} + \text{Adv}_{\mathcal{B}_3}^{\text{PRF}} \\ &\quad + C \frac{(Q_h + Q_s)^3}{2^{2\lambda}} + \frac{3Q_s}{2} \sqrt{\frac{Q_h}{2^\mu}} \end{aligned}$$

where μ is the min-entropy of the rounded aggregate commitment, and the parameters of the Hint-MLWE and SelfTargetMSIS instances are unchanged from the weak model.

Proof sketch. The reduction runs the simulator of the weak model. Its only new task is to answer queries of the last round in which the honest signers hold different views. It first computes the view classes, which it can do because it sees the inputs of every honest oracle call. For every class different from the all-honest class it samples the responses of the class uniformly, which is correct by Lemma 4.2; for the all-honest class it runs the simulator of the weak model verbatim. Freedom from the signer set is handled by noting that the message of the first round in Definition 3.1 does not depend on T and that all quantities that depend on T , namely $\lambda_{T,i}$ and the mask, are computed in the last round, so the simulator may defer its choice of T until then. No oracle point is reprogrammed that was not already reprogrammed in the weak model, so the loss terms and the parameters of the underlying problems are unchanged. The full proof is in Appendix B. $\square$

4.3 Necessary amount of masking

Every scheme in the Raccoon family spends the bulk of its online computation on masking. We show that the cost is $\Theta(t)$ full-entropy ring vectors, and that the constant achieved by prior work is twice the optimum. The following theorem addresses a lower bound on the amount of pseudorandom material per signer.

THEOREM 4.4. *Let Σ be a canonical linear-response scheme in which the masking term of signer i is computed non-interactively from keys shared with a subset $S_i \subseteq T \setminus \{i\}$ and public data, and suppose Σ is unforgeable against an adversary that corrupts any $t - 2$ members of a signer set of size t . Then $S_i = T \setminus \{i\}$ for every i , and the masking term of signer i must be an injective function of at least $|T| - 1$ values each of which is indistinguishable from uniform over $\mathcal{R}_q^\ell$. In particular each signer must generate at least $(t - 1)\ell\phi[\log_2 q]$ bits of pseudorandom material per signing session.*

Proof sketch. Suppose $h' \notin S_h$ for two indices h, h' and let the adversary corrupt $T \setminus \{h, h'\}$. Then every key used by h is shared with a corrupt party, so $\mathbf{m}_h$ is computable by the adversary, and the argument of Theorem 3.3 applied to h recovers $\lambda_{T,h}s_h$ from a single session because the adversary can choose two signer sets with distinct Lagrange coefficients. Hence $S_h = T \setminus \{h\}$. For the entropy claim, fix h and let the adversary corrupt $T \setminus \{h, h'\}$ for an arbitrary h' ; the only term of $\mathbf{m}_h$ that the adversary cannot compute is the one keyed by $\text{sd}_{\{h,h'\}}$, so that term alone must hide $c\lambda_{T,h}s_h$, which is uniform over $\mathcal{R}_q^\ell$. A single uniform pad over $\mathcal{R}_q^\ell$ is needed for each choice of h' , and the $t - 1$ pads must be independent because the adversary chooses h' after the seeds are fixed. Appendix C makes the reduction explicit. $\square$

The following statement addresses the optimality of the antisymmetric construction.

COROLLARY 4.5. *The antisymmetric masking as the values $\mathbf{m}_i = \sum_{j \in T \setminus \{i\}} \varepsilon_{ij} \text{PRF}(\text{sd}_{\{i,j\}}, \text{vw}_i)$ with one seed per unordered pair meets*

the bound of Theorem 4.4 with equality, whereas the ordered-pair construction of [4–6] uses $2t$ evaluations and $2n$ seeds per signer, twice the optimum in both.

PROOF. The antisymmetric construction evaluates the pseudo-random function once for each $j \in T \setminus \{i\}$, hence $t - 1$ times, which matches the bound of Theorem 4.4. Correctness of the zero-sum property holds as $\sum_{i \in T} \mathbf{m}_i = \sum_{\{i,j\} \subseteq T} (\varepsilon_{ij} + \varepsilon_{ji}) \text{PRF}(\text{sd}_{\{i,j\}}, \text{vw}) = 0$ when all views agree, and the secrecy requirement of Theorem 4.4 is met because for every pair of honest signers the shared seed is unknown to the adversary. The ordered construction stores $\text{sd}_{i,j}$ and $\text{sd}_{j,i}$ for every j and evaluates both, which doubles both quantities. $\square$

4.4 Lower bound for masked schemes

The same argument bounds the size of the message of the last round, and shows that Thimza is optimal in that component. The following proposition addresses the minimal size of a masked partial signature.

PROPOSITION 4.6. *Let Σ be a canonical linear-response scheme whose reconstruction coefficients γ_i are units of $\mathcal{R}_q$ that the adversary can vary over a set of size at least two, and let Σ be unforgeable against $t - 2$ corruptions. Then the message of the last round of an honest signer is uniform over $\mathcal{R}_q^\ell$ conditioned on the adversary's view, hence incompressible, and its length is at least $t\phi \log_2 q$ bits.*

PROOF. Fix an honest signer h and a corruption set of size $t - 2$ that leaves one further honest signer. By (2) the message is $\mathbf{z}_h = c\gamma_h s_h + f_h + \mathbf{m}_h$. By Theorem 4.4 the term of $\mathbf{m}_h$ keyed by the seed shared with the second honest signer is indistinguishable from uniform over $\mathcal{R}_q^\ell$ and independent of the remaining terms, so $\mathbf{z}_h$ is the sum of a uniform independent element and a quantity independent of it, hence uniform. A uniform element of $\mathcal{R}_q^\ell$ has entropy $\ell\phi \log_2 q$, so no encoding can represent it in fewer than $\ell\phi \log_2 q$ bits on average, and the bound follows. The hypothesis that the adversary can vary γ_h is what forces the mask to be present at all, as the attack of Section 3 shows; without it the response could be a short vector and could be compressed. $\square$

The message of the last round of Thimza is exactly $\ell\phi[\log_2 q]$ bits, so it meets Proposition 4.6 with equality. Together with the 32-byte first round and the truncated second round, this says that the only component of the transcript with room left is the second one, and Theorem 5.4 fixes how far it can be truncated.

As Figure 4 presents, on doubly logarithmic axes, one signer must generate a number of kibibytes of pseudorandom output that grows linearly in t ; the unit is one element of $\mathcal{R}_q^\ell$, which is 12 544 bytes at the parameters of Section 5.7. The upper curve is the rule of Threshold Raccoon, Ringtail and [5], which needs $2t$ such elements, the lower curve is the antisymmetric rule of Thimza, which needs $t - 1$, and the dotted line is the bound of Theorem 4.4, which the lower curve meets. The gap is a constant factor slightly above two at every t , and no asymptotic improvement is left in this component.

5 The Thimza lattice-based threshold signature

5.1 Design

Thimza signs with the Raccoon relation (1), so its output is a Raccoon signature and Definition 2.4 is met with $\Sigma' = \text{Raccoon}$. Signing has three rounds. The first is a hash commitment that binds the message but not the signer set, the second reveals the high-order

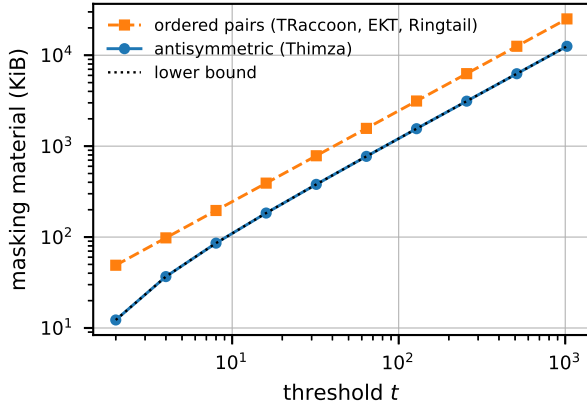


Figure 4: Pseudorandom material generated per signer and per signing session.

part of the commitment, and the third carries the masked response. Four elements distinguish it from Threshold Raccoon.

Commitment free of the signer set. The value of the first round is $\text{cmt}_i = H_0(\text{vk}, \text{msg}, i, \bar{\mathbf{w}}_i)$ and does not mention T , so a signer may commit before the signer set is known and the set may be finalised from the responders of the second round.

High-order commitment transmission. A signer publishes $\bar{\mathbf{w}}_i = \lfloor \mathbf{w}_i \rfloor_{v_c}$ rather than $\mathbf{w}_i$, dropping v_c least significant bits. The aggregate used by the protocol is $\mathbf{W} = \sum_{j \in T} 2^{v_c} \bar{\mathbf{w}}_j$, which differs from $\sum_j \mathbf{w}_j$ by a perturbation absorbed by the signature hint.

Antisymmetric view-derived masking. The value of the mask is $\mathbf{m}_i = \sum_{j \in T \setminus \{i\}} \varepsilon_{ij} \text{PRF}(\text{sd}_{\{i,j\}}, \text{vw})$ with a single seed per unordered pair and vw the digest of the whole second-round transcript. By Corollary 4.5 this is optimal, and by Lemma 4.2 it makes message authentication codes unnecessary. In particular no row mask is transmitted, since the masks already sum to zero.

No persistent store. By Corollary 3.6 a signer keeps only $(\mathbf{r}_i, \bar{\mathbf{w}}_i)$ between the first and the third round and erases it afterwards.

As Figure 5 presents, time runs downwards and each arrow is one broadcast, annotated with the object sent and its size at the level I parameters. The first arrow carries the 32 byte hash cmt_i and never mentions T , which is what frees the round from the signer set; the dashed box marks the work that does not depend on the message and can be done while the machine is idle, namely the sampling of $(\mathbf{r}_i, \mathbf{e}'_i)$ and the product $\mathbf{A}\mathbf{r}_i$. The second arrow carries the truncated opening $\bar{\mathbf{w}}_i$, a factor 3.8 below the corresponding object of Threshold Raccoon. The third carries the masked response, with the circle denoting the antisymmetric mask and its one edge per partner. The bar marked “erase” is where the token is destroyed, after which the signer holds no session state.

5.2 Interchangeability

A threshold scheme is useful as a drop-in replacement only if the verifier does not change [3]. We record that Thimza satisfies the strongest of the two notions usually distinguished, namely functional interchangeability with respect to verification, and that it does not satisfy the stronger notion of functional equivalence, which no probabilistic signature can satisfy once its randomness is produced jointly. The following theorem addresses the compatibility of Thimza signatures with the single-signer verifier.

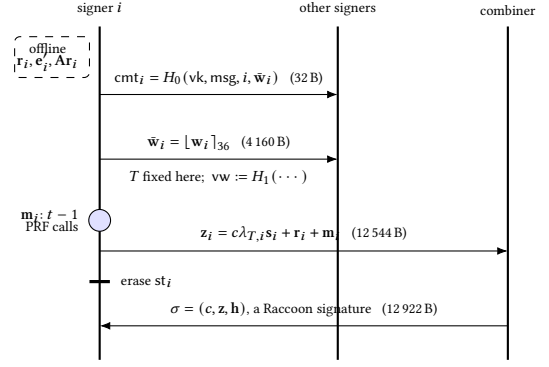


Figure 5: One execution of Thimza signing with level I sizes.

THEOREM 5.1. *Fix parameters as in Table 3 and let Σ' be the Raccoon signature scheme with the same parameters. For every $t \leq T_{\max}$, every signer set T with $|T| = t$ and every message, the output of Thimza.Combine on an honest execution is a triple $(c, \mathbf{z}, \mathbf{h})$ of the syntax of Σ' , lies in the support of Σ' .Sign(s, msg), and is accepted by Σ' . Verify without any transformation.*

PROOF. By Lemma 5.3 the combiner outputs $(c, \mathbf{z}, \mathbf{h})$ with $\mathbf{z} = c\mathbf{s} + \bar{\mathbf{r}}$, where $\bar{\mathbf{r}} = \sum_{j \in T} \mathbf{r}_j$, and with $\mathbf{h}$ defined by the same expression as in Σ' .Sign from that $\mathbf{z}$ and from the aggregate commitment. The three components therefore have the syntax of a Σ' signature, and Verify of Algorithm 3 is character for character Σ' . Verify, so acceptance is exactly the statement of Theorem 5.4.

It remains to show membership in the support. A single signer running Σ' .Sign samples $\mathbf{r} \leftarrow D_{\sigma_w \sqrt{T_{\max}}}^\ell$ and $\mathbf{e}' \leftarrow D_{\sigma_w \sqrt{T_{\max}}}^k$, computes $\mathbf{w} = \mathbf{A}\mathbf{r} + \mathbf{e}'$, sets $c = H_c(\text{vk}, \lfloor \mathbf{w} \rfloor_{v_w}, \text{msg})$, $\mathbf{z} = c\mathbf{s} + \mathbf{r}$ and $\mathbf{h} = \lfloor \mathbf{w} \rfloor_{v_w} - \lfloor \mathbf{A}\mathbf{z} - 2^{v_h} c\mathbf{t} \rfloor_{v_w}$. The support of Σ' .Sign(s, msg) is therefore the set of triples obtained in this way as $(\mathbf{r}, \mathbf{e}')$ ranges over the support of the two discrete Gaussians, which is all of $\mathcal{R}^\ell \times \mathcal{R}^k$. Given a Thimza output, put $\mathbf{r} := \bar{\mathbf{r}}$ and let $\mathbf{e}'$ be the unique vector with $\mathbf{A}\mathbf{r} + \mathbf{e}' \equiv \mathbf{W}$, which exists because the map is surjective in its second argument. These values lie in the support, they produce the same c , the same $\mathbf{z}$ and the same $\mathbf{h}$, so the triple is in the support of Σ' .Sign(s, msg). $\square$

Remark 5.2. Thimza is not functionally equivalent to Σ' , and no threshold scheme of this shape can be. Two effects separate the distributions. The aggregated randomness $\bar{\mathbf{r}}$ is a discrete Gaussian of parameter $\sigma_w \sqrt{t}$, which matches the single-signer parameter $\sigma_w \sqrt{T_{\max}}$ only at $t = T_{\max}$, and the hint additionally carries the truncation perturbation δ of Lemma 5.3, whose standard deviation per coefficient is $2^{v_c} \sqrt{t/12}$, that is 0.144 times the natural noise at every t . Equivalence is not needed for a deployment as a drop in replacement, since a relying party only runs the verifier, and it is the reason the standardization initiatives [3] separates the two notions.

5.3 Trust assumptions

It is worth noting in one place exactly what Thimza assumes, since the two contributions of Section 3 and Section 4 are precisely a reduction of that list. Unforgeability of Thimza requires: a trusted dealer that runs Algorithm 1 once, samples $(\mathbf{s}, \mathbf{e})$ from the prescribed Gaussians, erases them, and delivers to each party its share and its pairwise seeds over a private channel; a static adversary that

fixes a corruption set of at most $t - 1$ parties before key generation and thereafter sees the entire network; a broadcast medium that is neither authenticated nor reliable, so that the adversary may drop, reorder, duplicate and forge any message of any round, including messages that claim to come from an honest signer; honest parties that consume each token of the first round at most once and erase it, which Theorem 3.3 shows to be unavoidable and Corollary 3.6 shows to be sufficient; a quantum-accessible random oracle for H_0 , H_1 and H_c and a pseudorandom function for the masks; and the hardness of MLWE and SelfTargetMSIS at the parameters of Section 5.7 against a quantum adversary making at most Q_s signing queries.

Two assumptions that appear in the analyses of [4, 6] are absent from that list, namely authenticated channels between honest signers and knowledge of the signer set at the time of the first round. Two further assumptions are needed for liveness but not for unforgeability, namely a synchronous network and honest behaviour of the parties in the signer set; an adversary that controls the network can always make an execution fail, and Thimza, like every scheme of Table 1 except [15, 22], does not identify who caused the failure.

5.4 Deployment considerations

Three points matter for a deployment. First, the combiner is untrusted and may be one of the signers, since it only performs public arithmetic; if a dedicated combiner is used, the upload per signer is 16 736 bytes and the download is the same, independently of t . Second, the offline part of Round 1 is message-independent, so a signer may prepare a batch of tokens in advance; each token is $(\mathbf{r}_i, \bar{\mathbf{w}}_i)$, which is 12 544 + 4 160 bytes of local storage, and by Corollary 3.6 the batch needs no monotone counter, only a one-shot consumption flag. Third, the scheme inherits the masking-friendliness of Raccoon, since every operation of Algorithm 2 except the hash calls is linear over $\mathcal{R}_q$, so a side-channel-hardened implementation can share $\mathbf{s}_i$ and $\mathbf{r}_i$ additively without changing the protocol.

Algorithm 1 Thimza.KeyGen($1^\lambda, t, n$)

```

1: seedA  $\leftarrow \{0, 1\}^{2\lambda}$ ;  $\mathbf{A} := \text{Expand}(\text{seed}_A) \in \mathcal{R}_q^{k \times \ell}$ 
2:  $\mathbf{s} \leftarrow D_{\sigma_1}^\ell$ ,  $\mathbf{e} \leftarrow D_{\sigma_1}^k$ 
3:  $\mathbf{t} := \lfloor \mathbf{A}\mathbf{s} + \mathbf{e} \rfloor_{v_1}$ ;  $\text{vk} := (\text{seed}_A, \mathbf{t})$ 
4:  $\mathbf{p}_1, \dots, \mathbf{p}_{t-1} \leftarrow \mathcal{R}_q^\ell$ ;  $\mathbf{s}_i := \mathbf{s} + \sum_{u=1}^{t-1} \mathbf{p}_u \alpha_i^u$  for  $i \in [n]$ 
5: for each unordered pair  $\{i, j\} \subseteq [n]$  do
6:    $\text{sd}_{\{i,j\}} \leftarrow \{0, 1\}^\lambda$ 
7: end for
8:  $\text{sk}_i := (\mathbf{s}_i, (\text{sd}_{\{i,j\}})_{j \neq i}, \text{vk})$  for  $i \in [n]$ 
9: return  $(\text{vk}, (\text{sk}_i)_{i \in [n]})$ 

```

To go into the detailed procedure of Thimza, Algorithm 1 starts by expanding a short seed into the public matrix $\mathbf{A}$, so that the verification key stores 2λ bits rather than a full matrix. It then samples the short secret $\mathbf{s}$ and the short error $\mathbf{e}$ and publishes the key $\mathbf{t}$ with its v_1 least significant bits removed. Once the key is fixed, the secret is shared over $\mathcal{R}_q$ with a polynomial whose uniform coefficients are the $\mathbf{p}_u$ and whose evaluation point for party i is α_i , after which one seed is drawn for every unordered pair of parties, which is half of what earlier schemes store, and each signing key collects a share, the seeds of its own pairs and the verification key.

Algorithm 2 Thimza signing, code of signer i

```

Round 1. Sign1( $\text{sk}_i, \text{msg}$ )
1:  $\mathbf{r}_i \leftarrow D_{\sigma_w}^\ell$ ,  $\mathbf{e}'_i \leftarrow D_{\sigma_w}^k$ 
2:  $\mathbf{w}_i := \mathbf{A}\mathbf{r}_i + \mathbf{e}'_i$ ;  $\bar{\mathbf{w}}_i := \lfloor \mathbf{w}_i \rfloor_{v_c}$ 
3:  $\text{st}_i := (\mathbf{r}_i, \bar{\mathbf{w}}_i)$ ; return  $\text{cmt}_i := H_0(\text{vk}, \text{msg}, i, \bar{\mathbf{w}}_i)$ 

Round 2. Sign2( $\text{st}_i$ )
4: return  $\bar{\mathbf{w}}_i$ 

Round 3. Sign3( $\text{sk}_i, \text{st}_i, T, (\text{cmt}_j, \bar{\mathbf{w}}_j)_{j \in T}, \text{msg}$ )
5: if  $\exists j \in T : \text{cmt}_j \neq H_0(\text{vk}, \text{msg}, j, \bar{\mathbf{w}}_j)$  then abort
6:  $\text{vw} := H_1(\text{vk}, \text{msg}, T, (\text{cmt}_j)_{j \in T}, (\bar{\mathbf{w}}_j)_{j \in T})$ 
7:  $\mathbf{W} := \sum_{j \in T} 2^{v_c} \bar{\mathbf{w}}_j \bmod q$ ;  $\mathbf{w} := \lfloor \mathbf{W} \rfloor_{v_w}$ 
8:  $c := H_c(\text{vk}, \mathbf{w}, \text{msg}) \in \mathcal{C}$ 
9:  $\mathbf{m}_i := \sum_{j \in T \setminus \{i\}} \varepsilon_{ij} \text{PRF}(\text{sd}_{\{i,j\}}, \text{vw})$ 
10:  $\mathbf{z}_i := c \cdot \lambda_{T,i} \cdot \mathbf{s}_i + \mathbf{r}_i + \mathbf{m}_i \bmod q$ 
11: erase  $\text{st}_i$ ; return  $\mathbf{z}_i$ 

```

Algorithm 3 Thimza.Combine and Thimza.Verify

```

Combine( $\text{vk}, T, \text{msg}, (\bar{\mathbf{w}}_j, \mathbf{z}_j)_{j \in T}$ )
1: recompute  $\mathbf{w}$  and  $c$  as in Round 3
2:  $\mathbf{z} := \sum_{j \in T} \mathbf{z}_j \bmod q$ 
3:  $\mathbf{h} := \mathbf{w} - \lfloor \mathbf{A}\mathbf{z} - 2^{v_c} c \mathbf{t} \rfloor_{v_w} \bmod q_{v_w}$ 
4: return  $\sigma := (c, \mathbf{z}, \mathbf{h})$ 

Verify( $\text{vk}, \text{msg}, \sigma$ ) (identical to Raccoon)
5:  $\mathbf{w}' := \lfloor \mathbf{A}\mathbf{z} - 2^{v_c} c \mathbf{t} \rfloor_{v_w} + \mathbf{h}$ 
6: return  $[c = H_c(\text{vk}, \mathbf{w}', \text{msg})] \wedge [\|(\mathbf{z}, 2^{v_w} \mathbf{h})\|_2 \leq B]$ 

```

In Algorithm 2 a signer first samples the ephemeral randomness $\mathbf{r}_i$ and its noise $\mathbf{e}'_i$ with the flooding width σ_w , forms the commitment $\mathbf{w}_i = \mathbf{A}\mathbf{r}_i + \mathbf{e}'_i$, truncates it to $\bar{\mathbf{w}}_i$ by dropping v_c bits, and publishes only the 32 byte hash cmt_i of it, so this round names neither the signer set T nor any identifier. The truncated commitment is opened next. Once the openings arrive and the signer set is known, the signer checks every opening against its hash, hashes the whole transcript into the digest vw that keys the masks, rebuilds the aggregate $\mathbf{W}$ from the truncated openings and rounds it to $\mathbf{w}$, derives the challenge c from $\mathbf{w}$ and the message, forms the antisymmetric mask $\mathbf{m}_i$ from the $t - 1$ seeds it shares with the other signers, and returns the masked response $\mathbf{z}_i = c\lambda_{T,i}\mathbf{s}_i + \mathbf{r}_i + \mathbf{m}_i$, where $\lambda_{T,i}$ is the Lagrange coefficient. It then erases the token, which is what keeps the state bounded by the session.

Algorithm 3 adds the responses, at which point the masks cancel and $\mathbf{z}$ is a Raccoon response under the full key, and computes the hint $\mathbf{h}$ that repairs the two roundings. The verification algorithm is character for character the Raccoon verifier, which is the formal content of interchangeability.

5.5 Correctness

The following statement addresses the aggregation identity that underlies correctness.

LEMMA 5.3. *If all signers in T follow Algorithm 2 on a common view, then $\mathbf{z} = \sum_{j \in T} \mathbf{z}_j = \mathbf{c}\mathbf{s} + \sum_{j \in T} \mathbf{r}_j \bmod q$ and*

$$\mathbf{W} - (\mathbf{A}\mathbf{z} - 2^{v_c} c \mathbf{t}) = \sum_{j \in T} \mathbf{e}'_j + \delta + c\mathbf{e} - c\rho,$$

where $\delta = \sum_{j \in T} (2^{v_c} \bar{\mathbf{w}}_j - \mathbf{w}_j)$ is the truncation perturbation and $\rho = \mathbf{A}\mathbf{s} + \mathbf{e} - 2^{v_1} \mathbf{t}$ is the key rounding residue, with $\|\delta\|_\infty \leq t2^{v_c-1}$ and $\|\rho\|_\infty \leq 2^{v_1-1}$.

PROOF. By summing line 6 of Round 3 over $j \in T$ gives the value of $\mathbf{z} = c \sum_{j \in T} \lambda_{T,j} \mathbf{s}_j + \sum_{j \in T} \mathbf{r}_j + \sum_{j \in T} \mathbf{m}_j$. The first sum is $c\mathbf{s}$ by the reconstruction property of Shamir sharing, and the last sum vanishes because, all views being equal, each unordered pair contributes $\varepsilon_{ij} + \varepsilon_{ji} = 0$ times the same pseudorandom value. This proves the first identity. For the second, substitute $\mathbf{z}$ and use $2^u \mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e} - \boldsymbol{\rho}$:

$$\mathbf{A}\mathbf{z} - 2^u \mathbf{c}\mathbf{t} = c\mathbf{A}\mathbf{s} + \mathbf{A} \sum_j \mathbf{r}_j - c(\mathbf{A}\mathbf{s} + \mathbf{e} - \boldsymbol{\rho}) = \mathbf{A} \sum_j \mathbf{r}_j - c\mathbf{e} + c\boldsymbol{\rho}.$$

On the other hand $\mathbf{W} = \sum_j \mathbf{w}_j + \boldsymbol{\delta} = \mathbf{A} \sum_j \mathbf{r}_j + \sum_j \mathbf{e}'_j + \boldsymbol{\delta}$. Subtracting the two displays gives the claim. The bound on $\boldsymbol{\delta}$ holds because each summand is the rounding error of one coefficient, of magnitude at most 2^{v_c-1} , and the bound on $\boldsymbol{\rho}$ is the definition of rounding. $\square$

The following theorem addresses correctness of Thimza.

THEOREM 5.4. *Let $\tilde{\mathbf{e}}$ denote the right-hand side of the second identity of Lemma 5.3. If*

$$\left\| \left(\sum_{j \in T} \mathbf{r}_j, \tilde{\mathbf{e}} \right) \right\|_2 + 2^{v_w+1} \sqrt{k\varphi} \leq B, \quad (4)$$

then $\text{Verify}(\text{vk}, \text{msg}, \text{Combine}(\cdot)) = 1$. With σ_w, v_c, v_t, v_w and B chosen as in Section 5.7, inequality (4) holds except with probability at most $2^{-\lambda}$ for every $t \leq T_{\max}$, and in particular the choice $v_c = \lceil \log_2 \sigma_w \rceil - 1$ inflates the left-hand side by at most one per cent.

Proof sketch. The first component of the verification test is satisfied by construction: Combine defines $\mathbf{h}$ so that $\lfloor \mathbf{A}\mathbf{z} - 2^u \mathbf{c}\mathbf{t} \rfloor_{v_w} + \mathbf{h} = \mathbf{w}$, hence the recomputed challenge equals c . For the norm test, the rounding lemma of [4] bounds $\|2^{v_w} \mathbf{h}\|_2 \leq \|\tilde{\mathbf{e}}\|_2 + 2^{v_w+1} \sqrt{k\varphi}$, which gives (4). The tail bound follows from the concentration of a discrete Gaussian of parameter $\sigma_w \sqrt{t}$ in dimension $(\ell + k)\varphi$. The claim on v_c compares the standard deviations $2^{v_c} \sqrt{t/12}$ of $\boldsymbol{\delta}$ and $\sigma_w \sqrt{t}$ of $\sum_j \mathbf{e}'_j$, whose ratio is $2^{v_c} / (\sigma_w \sqrt{12})$ and is therefore independent of t . Appendix D contains the computation and Table 10 the numerical verification. $\square$

5.6 Security

Thimza satisfies TS-UF-CA unforgeability (Theorem 5.5), which implies TS-UF-0 (Proposition U.1) and transfers to the TS-UF-1 through TS-UF-4 notions (Proposition G.1), alongside correctness (Theorem 5.4) and interchangeability with the Raccoon verifier (Theorem 5.1). In this section, we address Thimza's unforgeability.

THEOREM 5.5. *In the quantum random oracle model, Thimza is unforgeable in the game of Definition 4.1 with session-bounded state under the MLWE and SelfTargetMSIS assumptions and the pseudorandomness of PRF against quantum distinguishers. Concretely, for every quantum adversary $\mathcal{A}$ that corrupts at most $t-1$ signers, makes at most Q_s classical signing queries and at most Q_h quantum superposition queries to H_0, H_1 and H_c , there are adversaries $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ running in essentially the time of $\mathcal{A}$ with*

$$\begin{aligned} \text{Adv}_{\text{Thimza}, \mathcal{A}}^{\text{ts-uf-ca}} &\leq \text{Adv}_{\mathcal{B}_1}^{\text{Hint-MLWE}_{q,k,\ell,Q_s,\sigma_t,\sigma_w,C}} \\ &+ \text{Adv}_{\mathcal{B}_2}^{\text{SelfTargetMSIS}_{q,k,\ell+1,C,\beta}} + \text{Adv}_{\mathcal{B}_3}^{\text{PRF}} + C \frac{(Q_h + Q_s)^3}{2^{2\lambda}} + \frac{3Q_s}{2} \sqrt{\frac{Q_h}{2^\mu}} \end{aligned}$$

where C is the absolute constant of the quantum collision bound of [36], $\beta = B + \sqrt{\omega} + (\omega 2^u + 2^{v_w+1}) \sqrt{k\varphi}$ and $\mu = k\varphi \log_2(q/2^{v_w}) - O(\lambda)$ is the min-entropy of the rounded aggregate commitment.

Proof sketch. The proof proceeds by a sequence of games in which the adversary keeps superposition access to H_0, H_1 and H_c throughout. Game 0 is the real game. Game 1 replaces every pseudorandom function evaluation under a seed shared by two honest signers with a uniform value, at the cost of Adv^{PRF} against a quantum distinguisher. Game 2 aborts on a collision of H_0, H_1 or H_c , which by the quantum collision bound of [36] costs $C(Q_h + Q_s)^3 / 2^{2\lambda}$ rather than the birthday term of the classical setting. Game 3 replaces the honest responses using Lemma 4.2: classes different from the all-honest class are answered with uniform values, and the all-honest class is answered by the single-honest-signer simulator that samples $\mathbf{z}$ first and reprograms H_c at the reconstructed commitment. Because the adversary queries in superposition, this step is not bounded by a union bound over query points; instead the reprogrammed point carries min-entropy μ from the unopened honest commitment, and the adaptive reprogramming lemma of [37] bounds the whole sequence of Q_s reprogrammings by $\frac{3Q_s}{2} \sqrt{Q_h 2^{-\mu}}$. Game 4 replaces $c\mathbf{s} + \sum \mathbf{r}_j$ by an independent Gaussian, which is a reduction to Hint-MLWE with Q_s hints and challenge distribution C ; the number of hints is the number of signing queries and not the number of hash queries, because the simulator reprograms the oracle only when a signing query is answered. In the final game the secret key is unused and a forgery yields a SelfTargetMSIS solution for $[\mathbf{A} | \mathbf{I}_k]$ with the stated bound, the extraction being the one of [32], which is stated directly against superposition queries and needs no rewinding. Appendix E gives the games and the reductions in full. $\square$

Remark 5.6. The number of Hint-MLWE hints in Theorem 5.5 is Q_s , the number of signing queries, and not the number of oracle queries. This is the point at which the profile with three rounds is better behaved than the two-round one, because the reduction of [6] must program the aggregation oracle at every query and therefore pays Hint-MLWE with Q_u hints, and the reduction of [9] pays the parameter inflation discussed in Section 1.1. Thimza pays neither, which is why its parameters are those of the driven by cryptanalysis regime. The dependence on Q_h that remains is the additive statistical loss of the collision and reprogramming steps, and it is the square-root loss of [37] rather than a linear one, which is why the min-entropy margin of Section 5.7 is ample.

Machine-Checked Security Proofs

We provide machine-checked proofs for security of Thimza in EasyCrypt [38], available at <https://anonymous.4open.science/r/thimza>, where Theorem 5.5 corresponds to lemma `tsufca_reduction` in `Security.ec`, and the proof model follows the game-hopping style of [4, 6]. Theorem 5.1 corresponds to lemma `verify_equiv` inside `Verify.ec`, and Theorem 3.3 corresponds to the of lemma `replay_recovers_share` inside `Statelessness.ec`, shown in Listing 1. Each assumption (MLWE, SelfTargetMSIS, Hint-MLWE, PRF) is a formal abstract module type paired with its security game in `Assumptions.ec` and `PRFSecurity.ec`, composed into the protocol in `Protocol.ec` and `Combine.ec`. Theorem 3.3 is demonstrated in EasyCrypt via a black-box adversary that replays a fixed token against two inconsistent views and recovers $y_h s_h$ by subtracting the two responses, exactly as in Appendix A, and Theorem 5.5 via a machine-checked seven-point triangle inequality summing the six per-hop bounds $|\Pr[G_i] - \Pr[G_{i+1}]|$ along the hybrid chain $G_0 \rightarrow \dots \rightarrow G_6$ in [?] code, with Lemma 4.2 verified separately as

Table 3: Thimza parameters. All sets use $\varphi = 512$, $\lceil \log_2 q \rceil = 49$, $\sigma_t = 2^{20}$, $T_{\max} = 1024$.

level	λ	Q_s	$\sigma_w \sqrt{T_{\max}}$	v_t	v_w	v_c	ℓ	k	ω
I	128	2^{60}	2^{42}	37	40	36	4	5	19
III	192	2^{64}	2^{42}	36	40	36	6	7	31
V	256	2^{60}	2^{41}	35	41	35	7	8	44

lemma `view_partition` in `ViewPartition.ec` and reused at the hop from G_3 to G_4 .

```
(* Singleton protocol matches plain signing *)
equiv singleton_is_plain (i0 : int) :
  SingletonSign.sign ~ PlainSign.sign :
  t{1} = t{2} /\ mu{1} = mu{2} /\ s0{1} = s0{2} /\ i{1} = i0 /\ tthr = 1
  ==> {res}.
proof.
proc.
seq 2 2 : (t{1} = t{2} /\ mu{1} = mu{2} /\ s0{1} = s0{2} /\ i{1} = i0
  /\ tthr = 1 /\ r{1} = r{2} /\ er{1} = er{2}).
+ by auto.
wp; skip => &1 &2 [#] ht hmu hso hi htt hr hr.
have e1 : shr s0{1} (fun _ => zerov) i{1} = s0{1}
  by exact (shr_share0 s0{1} (fun _ => zerov) i{1} htt).
have e2 : mask (singleton i{1}) witness i{1} = zerov
  by exact (singleton_mask_zero witness i{1}).
have ez : zcomb (singleton i{1})
  (fun j => (Rq.(*) (Hchal t{1} (rndw (mA ** r{1} + er{1}))) mu{1})) (
    lamSingle j))
  ** shr s0{1} (fun _ => zerov) j
  + (if j = i{1} then r{1} else zerov) + mask (singleton i{1})
  witness j)
  = (Hchal t{1} (rndw (mA ** r{1} + er{1}))) mu{1} ** s0{1} + r{1}.
+ rewrite /zcomb /singleton vmap_cons vmap_nil /= e1 e2 /lamSingle /=.
+ by smt (RqVec.Vector.ZModule.addr0).
by smt().
qed.
```

Listing 1: EasyCrypt statement and proof that Thimza restricted to a singleton committee computes exactly the ordinary Fiat–Shamir signature.

5.7 Parameters

Thimza inherits the module dimensions, modulus, widths and rounding amounts of Threshold Raccoon [4], so the instances of MLWE and SelfTargetMSIS, and hence the estimated bit security, are the same; the only new parameter is the truncation amount v_c . Table 3 lists the three levels. We re-derived the constraint system of [6] independently and verified that it reproduces their published parameters exactly, which validates the derivation script that also produces our numbers; the details are in Appendix F.

As Table 3 shows, each row is one NIST security level, Q_s is the number of signatures a key may produce, $\sigma_w \sqrt{T_{\max}}$ is the width of the aggregated ephemeral randomness and therefore the quantity that sets the signature size, v_t and v_w are the bits dropped from the verification key and from the aggregate commitment, v_c is the new number of bits dropped from the transmitted commitment share, ℓ and k are the module dimensions of the secret and of the samples, and ω is the Hamming weight of the challenge. The value $v_c = 36$ at level I is the largest for which Theorem 5.4 costs at most one per cent of the verification bound, and it shrinks the message of the second round from 15 680 to 4 160 bytes.

5.8 Choice of the parameter sets

Three constraints fix the sets of Table 3. The flooding width must satisfy $\sigma_w \geq \eta_\varepsilon(\mathbb{Z}^\varphi) \omega \sqrt{Q_s}$ so that the reduction to Hint-MLWE of Theorem 5.5 applies with Q_s hints of challenge weight ω ; at level I with $Q_s = 2^{60}$ and $\omega = 19$ this gives $\sigma_w \geq 2^{35.8}$ and we take 2^{37} . The

modulus must exceed the aggregated response by enough room for the two roundings, namely $q \geq 2^6 \sigma_w \sqrt{T_{\max}}$, which gives $\lceil \log_2 q \rceil = 49$. The key rounding v_t is the largest value for which $\omega 2^{v_t} \sqrt{k\varphi}$ stays below the aggregated noise, which gives 37 at level I and decreases with the level because ω grows. The new parameter v_c is then set by Theorem 5.4. Because these are exactly the constraints of [4] plus one, the MLWE instance $(A, [As + e]_q)$ and the SelfTargetMSIS instance of Theorem 5.5 are the ones already analysed there, and the estimated hardness is unchanged; Appendix I states the estimator² we use and validates it.

6 Complexity analysis

Before measuring the schemes we compare them analytically. We count operations in $\mathcal{R}_q$, distinguishing a number theoretic transform of one ring element, which costs $\Theta(\varphi \log \varphi)$ integer operations, from a coefficientwise product or a pseudorandom expansion, which costs $\Theta(\varphi)$. We write t for the threshold, ρ for the number of columns of the matrix of the first round of a scheme with two rounds, namely $\rho = \bar{d} + 1$ for Ringtail and $\rho = \text{rep}$ for the scheme of [5], and $L = \lceil \log_2 q \rceil$.

Time. In all of the schemes the key generation of a dealer costs $\Theta(n(k + \ell)\varphi \log \varphi)$ because it evaluates the sharing polynomial at n points. The signing cost splits into a commitment part and a response part. The commitment part of Thimza and Threshold Raccoon is a single matrix-vector product, $\Theta(k\ell\varphi \log \varphi)$; for a scheme with two rounds it is a matrix-matrix product of width ρ , hence $\Theta(k\ell\rho\varphi \log \varphi)$, and Ringtail additionally performs a rank test on the aggregate, which costs $\Theta(\varphi k^2 \bar{d})$ field operations. The response part is dominated by masking and costs $\Theta(\kappa_m t \ell \varphi)$ where $\kappa_m = 1$ for Thimza by Corollary 4.5 and $\kappa_m = 2$ for all prior schemes; Threshold Raccoon further pays $\Theta(t)$ symmetric authentications. Summing, the signing time per signer of Thimza is

$$\Theta(k\ell\varphi \log \varphi + t\ell\varphi),$$

against the $\Theta(k\ell\varphi \log \varphi + 3t\ell\varphi)$ for the Threshold Raccoon, and $\Theta(k\ell\rho\varphi \log \varphi + \varphi k^2 \bar{d} + 2t\ell\varphi)$ for Ringtail and $\Theta(k\ell\rho\varphi \log \varphi + 2t\ell\varphi)$ for the scheme of [5]. Since $\rho = 49$ for Ringtail and $\rho = 15$ for [5], the commitment term of those schemes exceeds that of Thimza by one to two orders of magnitude, and the rank test adds a term that is quadratic in k .

Space. Communication per signer is $32 + k\varphi(L - v_c) + \ell\varphi L$ bits for Thimza, $32 + 2\ell\varphi L + k\varphi L + 16t \cdot 8$ bits for Threshold Raccoon, $k\rho\varphi L + \ell\varphi L + 16t \cdot 8$ bits for Ringtail and $k\rho\varphi L + \ell\varphi L$ bits for the scheme of [5]. Three differences are tangible. First, Thimza removes the transmitted row mask, which alone is $\ell\varphi L$ bits. Second, it truncates the commitment share, which removes $k\varphi v_c$ bits. Third, it removes the authentication tags of $128t$ bits, which is the only term in prior schemes with three rounds and in those of the Ringtail type that grows with t ; Thimza is therefore the only scheme in the family whose communication is exactly independent of the threshold.

As Table 4 shows, the rows are the procedures of Definition 2.4 followed by the four size measures and the state, with n the number of key holders, t the threshold, φ the ring degree, (k, ℓ) the module dimensions, $L = \lceil \log_2 q \rceil$, ρ the width of the matrix of the first round in a scheme with two rounds, $\bar{d} = \rho - 1$, the three v values the rounding amounts, and $\sigma_z = \sigma_w \sqrt{T_{\max}}$ the width of the aggregated response; a dash marks a round that does not exist. Three facts follow. The row for `Sign1` carries the factor ρ that a preprocessable

²<https://github.com/malb/lattice-estimator>

Table 4: Asymptotic cost of the main procedures. Time is counted in $\mathcal{R}_q$ operations for one signer, space in bits sent by one signer.

Procedure	Thimza (Ours)	TRaccoon [4]	EKT [5]	Ringtail [6]
KeyGen	$n(k+t)\varphi \log \varphi$	$n(k+t)\varphi \log \varphi$	$n(k+t)\varphi \log \varphi$	$n(k+t)\varphi \log \varphi$
Sign ₁	$kt\varphi \log \varphi$	$kt\varphi \log \varphi + t\ell\varphi$	$kt\varphi \log \varphi$	$kt\varphi \log \varphi$
Sign ₂	0	t	$k\rho\varphi \log \varphi + 2t\ell\varphi$	$k\rho\varphi \log \varphi + \varphi k^2 d + 2t\ell\varphi$
Sign ₃	$t\ell\varphi + kt\varphi \log \varphi$	$2t\ell\varphi + kt\varphi \log \varphi$	—	—
Combine	$t\ell\varphi + kt\varphi \log \varphi$	$2t\ell\varphi + kt\varphi \log \varphi$	$t\ell\varphi + kt\varphi \log \varphi$	$t\ell\varphi + kt\varphi \log \varphi$
Verify	$kt\varphi \log \varphi$	$kt\varphi \log \varphi$	$kt\varphi \log \varphi$	$kt\varphi \log \varphi$
$ tr /t$	$k\varphi(L - v_c) + \ell\varphi L$	$2\ell\varphi L + k\varphi L + 128t$	$k\rho\varphi L + \ell\varphi L$	$k\rho\varphi L + \ell\varphi L + 128t$
$ \sigma $	$\ell\varphi \log \sigma_z + k\varphi \log \frac{\sigma_z}{2^{1/W}}$	same	same	same
$ vk $	$k\varphi(L - v_c) + 2\lambda$	same	same	same
$ sk_i $	$\ell\varphi L + \lambda(n-1)$	$\ell\varphi L + 2\lambda n$	$\ell\varphi L + 2\lambda n$	$\ell\varphi L + 2\lambda n$
state	$O(1)$ token	$O(Q_s)$ ids	$O(1)$ token	$O(1)$ token

Table 5: Modelled $\mathcal{R}_q$ -operations per signer at level I, in units of 10^3 .

Scheme	commit	aggregate	per partner	$t=64$	$t=1024$
Thimza (Ours)	92	92	$4.61t$	479	4 904
TRaccoon	92	92	$6.69t$	612	7 034
TalonG	23	68	$10.2t$	746	10 577
Hermine	958	240	$26.6t$	2 901	—
EKT	3 041	338	$46.8t$	6 374	51 302
Ringtail	5 620	1 589	$104t$	13 865	113 705

first round costs. The rows for Sign₂ and Sign₃ show that Thimza halves the masking coefficient and drops the authentication term, while Hermine drops the masking entirely and pays instead a factor rep in its first round and a share whose size grows with n . The communication row shows that Thimza is the only entry with no term in t , and that its leading term is below that of Threshold Raccoon by $\ell\varphi L + k\varphi v_c$ bits, which is 70 per cent at the level I parameters.

Numeric instantiation. Substituting the level I parameters of each scheme into Table 4 gives the counts of Table 5, which predict the measurements of Section 7 without any calibration.

As Table 5 shows, “commit” counts the object of the first round, one matrix vector product for the schemes with three rounds and a matrix matrix product of width ρ or rep for the others; “aggregate” counts the fixed cost of forming the challenge, including the rank test of Ringtail; and “per partner” is the coefficient of t , which collects the masks and the summation of what the other signers send. The predicted ratios to Thimza at $t = 64$ are 1.3, 1.6, 6.1, 13.3 and 28.9 for Threshold Raccoon, TalonG, Hermine, the scheme of [5] and Ringtail, against measured 1.3, 2.2, 6.0, 9.5 and 24.8 in Table 8; the model therefore predicts both the ordering and the magnitudes without calibration.

As Figure 6 presents, the models of Table 4 instantiated with each scheme’s own parameters are eventually linear in t , and they differ in the intercept, set by the commitment, and in the slope, set by the masking coefficient and by the width of the object each signer must sum. The curve of Thimza is the lowest at every t , below $t \approx 32$ because its commitment is one matrix vector product and above it because it generates $t - 1$ rather than $2t$ pseudorandom vectors. The curve of Hermine stops at $t = 64$, the largest committee for which its parameters are given, and lies above Thimza throughout because of the factor rep = 13 in its first round.

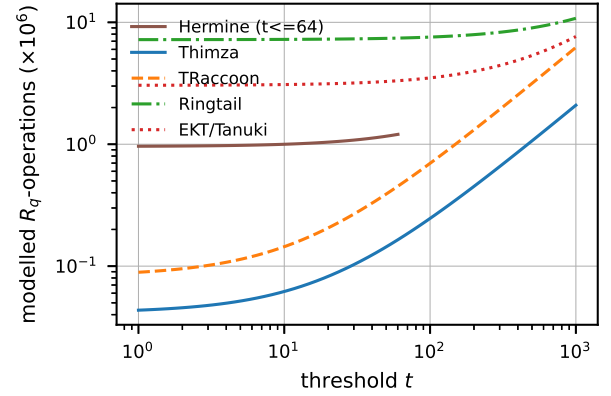


Figure 6: Modelled signing cost per signer from Table 4.

7 Implementation and practical evaluation

Here, we go through our experiments, where we observe that Thimza is the most time-efficient lattice-based threshold signature and the most space-efficient among the interchangeable schemes.

7.1 Setup

We deploy Thimza, Threshold Raccoon [4], the scheme with two rounds of [5] in the form specified by its standardization writeup, Ringtail [6] and TalonG [8] in one framework in <https://anonymous.4open.science/r/thimza> that shares one arithmetic backend, one sampler, one hash and one serializer, so that the reported differences are attributable to the protocols. The backend performs negacyclic number theoretic transforms over $\mathbb{Z}_q$ for $q < 2^{51}$ with a floating point quotient and 64-bit wrap around reduction, and it is validated against schoolbook multiplication. Every scheme is validated for correctness on all of $(t, n) \in \{(2, 3), (3, 5), (4, 6), (5, 8), (7, 10)\}$, and Thimza signatures are additionally checked against an independent Raccoon verifier, which is the operational content of interchangeability. Each measurement is the minimum of three executions, and our experiments are conducted on an Apple MacBook Air (2022) with the M2 chip, A15 Bionic (64-bit ARM-based), an 8-core CPU featuring a base clock speed of 3.49 GHz, and 16GB of memory, Tahoe OS 26.2 build 25C56, CPython 3.12.3 with NumPy 2.3.1.

7.2 Space

Table 6 shows the sizes produced by our serializer. Our model reproduces the published numerical results of the prior schemes closely, which supports its use for Thimza: it gives 40 848 bytes for Threshold Raccoon against the published 40 800, 602 112 bytes for the Ringtail preprocessing message against the published 602 112, and 14 400 bytes for the online message of [5] against the published 14 400.

As Table 6 shows, every byte count comes from the same serializer, so the entries compare directly; $|vk|$ is a 32 byte seed for A together with the rounded key t , “round 1” is the first protocol message, which for Thimza is one hash and for the schemes with two rounds is the whole preprocessing object, and the last block repeats the comparison at $t = 64$, the largest committee for which Hermine gives parameters. We observe three points here. Thimza and Threshold Raccoon share a verification key and a signature because they share the lattice parameters, so the factor 3.42 between them comes from the protocol alone. Thimza sends less than

Table 6: Sizes in bytes at the 128 bit level, from our implementation. The last two columns are the threshold at which the row is evaluated and the total. The TalonG row uses rounding where the original uses modulus switching, so its published total is smaller; Table 1 quotes the published values.

Scheme	$ vk $	$ \sigma $	round 1	later rounds	t	total
Thimza	3 872	13 152	32	16 704	1024	16 736
TRaccoon [4]	3 872	13 152	12 576	44 608	1024	57 184
TalonG [8]	1 312	21 536	11 008	35 840	1024	46 848
EKT [5]	4 256	14 176	264 000	14 400	1024	278 400
Ringtail [6]	4 640	14 208	602 112	27 136	1024	629 248
Thimza	3 872	13 152	32	16 704	64	16 736
Hermine [15]	2 848	12 832	53 248	20 992	64	74 240

Table 7: Sizes in bytes at all three security levels and $t = 1024$. The row marked * is evaluated at $t = 64$, the largest committee for which its developers give parameters.

Level	Scheme	$ vk $	$ \sigma $	total comm.
I	Thimza	3 872	13 152	16 736
	TRaccoon	3 872	13 152	57 184
	EKT	4 256	14 176	278 400
	Ringtail	4 640	14 208	629 248
	TalonG	1 312	21 536	46 848
	Hermine*	2 848	12 832	74 240
III	Thimza	5 856	19 552	24 672
	TRaccoon	5 856	19 552	76 000
	EKT	7 200	18 784	355 200
	Ringtail	6 560	21 408	790 656
V	Thimza	7 200	21 280	29 152
	TRaccoon	7 200	21 280	85 408
	EKT	10 272	23 392	577 728
	Ringtail	8 736	30 304	1 242 112
	TalonG	2 080	43 040	94 720

any other scheme, a factor 1.65 below the published 27 546 bytes of TalonG and a factor 4.44 below Hermine. The signature of TalonG is the largest because it carries the commitment opening, which is also why it is not interchangeable, while the signature and the key of Hermine are the smallest, at the cost of a first round that is 1660 times larger than ours.

As Table 7 shows, the three size measures are repeated at the three NIST security levels with each scheme’s own parameter set; TalonG has no published level III set and Hermine no set beyond level I in our implementation, which is why those rows are absent. The ratio between Thimza and Threshold Raccoon is 3.42, 3.08 and 2.93, so the saving is not an artefact of one parameter set, and it softens at the higher levels because the module dimensions grow and the response of the third round, which Thimza does not shrink, takes a larger share. The ratio to Ringtail is 37.6, 32.0 and 42.6 and stays above thirty throughout, because the trapdoor width grows with the module dimension while our transcript grows only with the response.

As Figure 7a presents, communication per signer against t on logarithmic axes, the curves of Threshold Raccoon and of Ringtail rise with t because of their $128t$ bits of authentication tags, while those of Thimza, Hermine and [5] are flat, since none of them transmits data per partner; Thimza is the lowest curve at every t

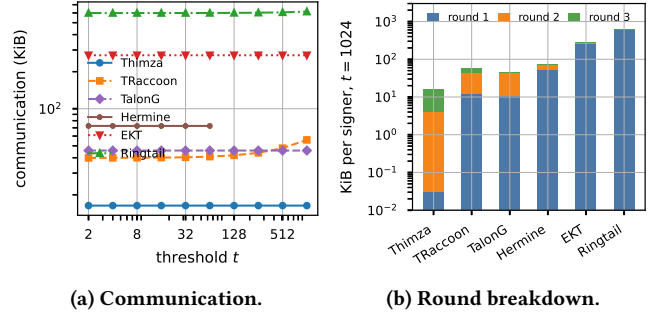


Figure 7: Communication per signer, as a function of t and split by round at $t = 1024$.

Table 8: Signing time per signer in milliseconds, single core, one framework.

	t	Thimza	TRaccoon	Hermine	TalonG	EKT	Ringtail
	4	2.22	2.44	17.87	4.79	23.53	78.97
	16	3.20	3.86	21.85	6.80	32.19	97.45
	64	6.82	9.07	40.96	14.97	64.47	169.12
	256	22.81	32.64	—	47.57	191.54	449.10
	1024	78.29	122.00	—	179.20	710.00	1584.43
ratio at $t = 64$		—	1.33	6.01	2.20	9.45	24.80
ratio at $t = 1024$		—	1.56	—	2.29	9.07	20.24

and the gap widens. Figure 7b splits the same quantity into rounds. The first round of Thimza is the 32 byte commitment, invisible at this scale, and its bulk is the response of the third round, which Proposition 4.6 shows to be minimal for a masked partial signature. For the schemes with two rounds almost the whole bar is the first round, so their preprocessing, and not their online message, is what fills the bandwidth budget.

7.3 Time

Table 8 shows the signing time per signer. Thimza is the fastest at every threshold, and the advantage grows with t because of the halved masking coefficient.

As Table 8 shows, entries are milliseconds spent by one signer on its whole share of the protocol, excluding the network, and the last two rows divide each entry by that of Thimza. Every column grows linearly in t for large t , which is the term predicted by Table 4; the slopes differ by the number of pseudorandom vectors a signer generates and by the width of the object it must sum. The intercepts differ by the width of the object of the first round, namely 49 ring columns for Ringtail, 15 for [5] and 13 for Hermine against a single vector for Thimza, which is why those three columns start an order of magnitude higher. Thimza is the fastest among threshold scheme.

As Figure 8a presents, every curve of Table 8 becomes a straight line of slope one for large t , which confirms the linear term, and the vertical offsets are the constants analysed above; the curve of Thimza lies below all others over the whole range. Figure 8b plots signature size against communication, both logarithmic, with the six implemented schemes as filled markers and the remaining published ones as grey crosses; a point that is down and to the left dominates. Thimza is the leftmost point of the whole plot, and among the schemes that reach $t = 1024$ it is also the lowest except for [5], whose signature is smaller only under a non-standard

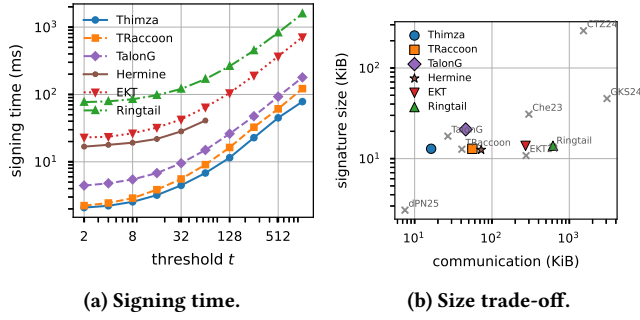


Figure 8: Measured signing time and the size trade-off at $t = 1024$.

Table 9: Signing time per signer split by round, in milliseconds.

Scheme	t	round 1	round 2	round 3	total
Thimza	1024	1.19	0.00	77.10	78.29
TRaccoon	1024	40.23	1.20	80.57	122.00
TalonG	1024	2.32	176.88	—	179.20
EKT	1024	10.79	699.21	—	710.00
Ringtail	1024	25.43	1559.00	—	1584.43
Thimza	64	1.06	0.00	5.76	6.82
Hermine	64	6.14	34.82	—	40.96

assumption. Hermine and the compact scheme of [26] sit below and to the right, which is the trade they make because a smaller signature bought with more bandwidth and a smaller committee.

7.4 Round Times

Table 9 splits the signing time of the scheme with three rounds by round and shows the scheme with two rounds for reference. The comparison isolates the two mechanisms of Section 4: the second round of Thimza is free because it only forwards an already computed value, whereas the second round of Threshold Raccoon computes t authentication tags, and the third round of Thimza generates half the pseudorandom material.

As Table 9 shows, a dash marks a round that does not exist. The second round of Thimza returns a value already computed, so it is free, while the first round of Threshold Raccoon costs 40 milliseconds against 1.2 for Thimza because it also builds and publishes the row mask, an object of 12 544 bytes that the antisymmetric construction makes unnecessary. The third rounds are almost equal, at 77.1 against 80.6, since Threshold Raccoon has already spent half of its masking budget in the first round; summing the three rounds gives the factor of Corollary 4.5. For the schemes with two rounds the whole cost sits in the online round, which combines the aggregation of a wide object with the masking, and for Hermine it sits there as well even though no mask is generated, because the object to be summed has $\text{rep} = 13$ columns.

7.5 Latency in the wide area network

Round complexity and bandwidth pull in opposite directions, so we combine them in one model. Using the round trip time of 170 milliseconds and the throughput of 183 megabits per second measured in the eight region deployment of [6], the end to end time of a signing execution is $\rho_{\text{on}} \cdot \text{RTT} + \text{bytes/throughput} + \text{compute}$.

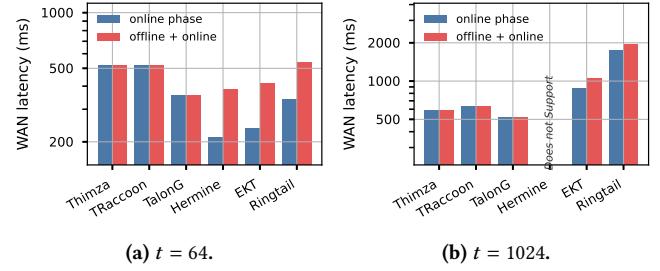


Figure 9: Modelled wide area latency from the network measurements of [6], at both committee sizes.

As Figure 9 shows, the left bar of each pair is the part of an execution that depends on the message and the right bar is a complete execution including preprocessing; we plot both committee sizes because they rank the schemes differently, and Hermine has no parameter set at the larger one, which the panel marks in place of its bars. At $t = 64$ the round trips dominate every other term and the ranking follows the number of rounds that depend on the message as Hermine, the scheme of [5] and Ringtail lead with one such round at 212, 235 and 340 milliseconds, TalonG follows with two at 357, and Thimza pays three for 518. At $t = 1024$ the ordering inverts, because the compute term is the only one that grows with the committee as a complete execution costs 521 milliseconds for TalonG and 589 for Thimza, against 634 for Threshold Raccoon, 1062 for the scheme of [5] and 1952 for Ringtail. Round complexity buys latency at small committees, and computation and bandwidth buy it at large ones, which is where Thimza is placed.

Limitations

Although Thimza is generally more efficient than other schemes from both time and space cost, it has a few small limitations in narrow scenarios. As Figure 9 shows, although Thimza sends less than every scheme we compare against and signs faster than all of them at $t = 1024$, its three rounds all depend on the message, so where the round trips dominate the schemes that preprocess a round lead on latency: at $t = 64$ a complete execution costs Thimza 518 milliseconds against 357 for TalonG, 384 for Hermine and 417 for the scheme of [5], which is an ordering that reverses only once the committee grows. On size, as Table 1 shows, Thimza is behind on two measures and ahead on the rest, the verification key of TalonG at 1 312 bytes against our 3 872 and the key and the signature of Hermine at 2 848 and 12 832 against our 3 872 and 13 152; the scheme of [5] shows a shorter signature under the non-standard AOM-MLWE assumption, and both of these smaller objects are bought with first rounds of 11 008 and 53 248 bytes against our 32.

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A Proof of Theorem 3.3

We restate the setting. Let Σ be canonical linear-response with $\rho \geq 2$ rounds, let the randomness of the first round ρ_h of signer h be sampled in round 1, and let the algorithm for partial signing of the last round be a deterministic function $\text{Rsp}(\text{sk}_h, \rho_h, \text{vw})$, which is the content of Definition 3.2 once the token ρ_h is regarded as part of the input rather than as a consumable resource.

PROOF OF THEOREM 3.3. The adversary $\mathcal{A}$ proceeds as follows. It corrupts the set C of size $t - 1$, chosen so that $T := C \cup \{h\}$ has size t for an honest index h ; such a choice exists because $t \leq n$ leaves at least $n - t + 1 \geq 1$ honest indices. It queries the oracle of the first round of h once and obtains the commitment of h . It then assembles two vectors of messages of the first round of the corrupt parties, $\mathbf{u}$ and $\mathbf{u}'$, that agree on the signer set T and on the message, and differ in the commitment of one corrupt party. Let $\text{vw} = (T, \text{msg}, \mathbf{u}, \text{cmt}_h)$ and $\text{vw}' = (T, \text{msg}, \mathbf{u}', \text{cmt}_h)$ be the two resulting views. It queries the oracle of the last round of h twice, on vw and on vw' ; statelessness guarantees that both queries are answered and that both use the same token ρ_h .

Write $c = \text{Chl}(\text{vw})$ and $c' = \text{Chl}(\text{vw}')$. Since the challenge function is a quantum-accessible random oracle applied to the aggregate commitment, and the two aggregates differ, c and c' are independent and uniform in C ; in particular $c \neq c'$ except with probability $1/|C| \leq 2^{-\lambda}$. Since T is the same in both views, $\gamma_h(\text{vw}) = \gamma_h(\text{vw}') =: \gamma$. By (2),

$$z_h - z'_h = (c - c')\gamma s_h + (f_h(\rho_h, \text{vw}) - f_h(\rho_h, \text{vw}')) + (\mathbf{m}_h(\text{vw}) - \mathbf{m}_h(\text{vw}')).$$

We treat the last two brackets in turn. For the third bracket, h is the unique honest member of T , so every seed used by h is shared with a

corrupt party and is therefore in the adversary's possession. Hence $\mathcal{A}$ computes $\mathbf{m}_h(\mathbf{vw})$ and $\mathbf{m}_h(\mathbf{vw}')$ itself. This also covers masking mechanisms that are not view-derived, for instance the identifier-derived mechanism of [4], since in that case the two masks are simply equal and the bracket vanishes.

For the second bracket, we distinguish two cases. If f_h does not depend on the view, which is the situation of a scheme with three rounds where $f_h(\rho_h, \mathbf{vw}) = \mathbf{r}_h$, the bracket vanishes. If f_h depends on the view through a public transform, which is the situation of a scheme with two rounds where $f_h(\rho_h, \mathbf{vw}) = \mathbf{R}_h \mathbf{u}(\mathbf{vw})$ for a public aggregation vector $\mathbf{u}(\mathbf{vw})$, then $\mathcal{A}$ repeats the experiment $\ell\varphi$ times with the same token and pairwise distinct views. Each repetition yields a linear equation over $\mathcal{R}_q$ in the unknowns $(y_{s_h}, \mathbf{R}_h)$ whose coefficient vector is $(c - c', \mathbf{u}(\mathbf{vw}) - \mathbf{u}(\mathbf{vw}'))$; these coefficient vectors are independent and uniform, so after $\rho + 1$ repetitions the linear system has full rank except with probability at most φ/q by the Schwartz–Zippel bound applied in each of the φ number theoretic transform slots, and $\mathcal{A}$ solves it. Statelessness is what permits the repetitions, since a signer with session-bounded state would have erased the token after the first of them.

In both cases $\mathcal{A}$ obtains $(c - c')y_{s_h}$ and, since $c - c'$ has ℓ_1 norm at most 2ω and is a difference of two independent uniform elements of C , it is invertible in $\mathcal{R}_q$ except with probability at most φ/q over the choice of the random oracle outputs, because $\mathcal{R}_q$ splits into φ fields and a fixed nonzero element fails to be invertible only if one of its number theoretic transform coordinates vanishes. Multiplying by the inverse yields y_{s_h} .

Finally, from $y_{s_h} = \lambda_{T,h}s_h$ and the knowledge of $\lambda_{T,h}$, which is public, $\mathcal{A}$ obtains s_h , hence together with the $t - 1$ corrupt shares the whole secret s by Lagrange interpolation, and it signs any message of its choice. The probability that any of the two failure events occurs is at most $2^{-\lambda} + \varphi/q$, which proves the claim. $\square$

Remark A.1. The theorem does not contradict the existence of stateless threshold signatures altogether. Its hypothesis is that the randomness of the first round is committed before the view that determines the challenge is complete, which is the defining feature of an interactive Fiat–Shamir protocol. Round-optimal constructions from threshold fully homomorphic encryption [16, 17] evade it because their partial signature is a deterministic function of the key and the message, with no separate commitment round, and they are indeed stateless. The price is the homomorphic evaluation of the signing circuit.

B Proof of Theorem 4.3

PROOF OF THEOREM 4.3. Let $\mathcal{A}$ be an adversary in the game of Definition 4.1. We construct an adversary $\mathcal{A}'$ in the weak game and show that the two advantages differ by at most $\text{Adv}_{\mathcal{B}_3}^{\text{PRF}}$.

$\mathcal{A}'$ runs $\mathcal{A}$ and forwards its key generation and corruption phase unchanged. It maintains for every open session the list of the queries of the last round made so far. When $\mathcal{A}$ makes a query of the last round to honest signer i with a vector of messages of the first round, $\mathcal{A}'$ computes the view $\mathbf{vw}_i$ that i would compute and records the pair $(i, \mathbf{vw}_i)$.

We argue by a hybrid over the sessions. Fix a session and let H be its honest members. Consider the partition $H/\sim$ into view classes. Let S^* be the class equal to H if it exists, and undefined otherwise.

Hybrid 0. The real game of Definition 4.1.

Hybrid 1. Identical, except that every evaluation $\text{PRF}(\text{sd}_{\{i,j\}}, \cdot)$ with i, j both honest is replaced by a lazily sampled uniform function. A standard hybrid over the at most $\binom{|H|}{2}$ such seeds bounds the distance by $\text{Adv}_{\mathcal{B}_3}^{\text{PRF}}$.

Hybrid 2. Identical, except that for every class $S_a \neq S^*$ the responses $(z_i)_{i \in S_a}$ are replaced by uniform elements of $\mathcal{R}_q^\ell$ subject to no constraint. We claim Hybrid 2 and Hybrid 1 are identically distributed. Order the classes arbitrarily and consider $S_a \neq S^*$. As in the proof of Lemma 4.2 choose $h \in S_a$ and $h' \in H \setminus S_a$ and consider the value $v = \varepsilon_{hh'} F_{\{h,h'\}}(\mathbf{vw}_h)$, where F is the uniform function introduced in Hybrid 1. The point $\mathbf{vw}_h$ is not queried by any other signer, since only h and h' hold the key $\text{sd}_{\{h,h'\}}$ and h' queries $\mathbf{vw}_{h'} \neq \mathbf{vw}_h$. Therefore v is uniform and independent of everything else in the experiment, and it occurs additively in z_h and in no other response. Consequently z_h is uniform and independent of the remaining responses; repeating the argument for the remaining members of S_a , using a different partner h' each time if necessary and otherwise using the freedom already established, shows that the whole vector $(z_i)_{i \in S_a}$ is uniform. The claim follows.

Hybrid 3. Identical, except that the responses of S^* , when it exists, are produced by the simulator of the weak model. By Lemma 4.2 the joint distribution of $(z_i)_{i \in S^*}$ in Hybrid 2 is exactly the distribution the simulator of the weak model is designed to produce, because all members of S^* hold the same view and the residual mask terms are the ones keyed with corrupt seeds, which the simulator can compute. Hence Hybrid 3 and Hybrid 2 are identical.

In Hybrid 3 every honest response is produced either uniformly or by the simulator of the weak model, so $\mathcal{A}'$ can run $\mathcal{A}$ using only its own weak-model oracles, because it forwards to its oracles exactly the queries of the class S^* , which are consistent in their views by construction and therefore admissible in the weak game, and it answers all other queries itself with uniform values.

It remains to handle coalition agnosticism. In the weak game the oracle of the first round takes T ; in the game of Definition 4.1 it does not. By Definition 3.1 the message of the first round is $\text{Cmt}(\text{sk}_i; \rho_i)$ and does not depend on T , and the two quantities of the last round that do depend on T , namely $y_i = \lambda_{T,i}$ and $\mathbf{m}_i$, are computed in the last round. Therefore $\mathcal{A}'$ may postpone its own oracle of the first round call until $\mathcal{A}$ discloses T in the last round, and answer $\mathcal{A}$'s query of the first round with the value it will later obtain; to do so it needs the message of the first round before it calls its oracle, which it obtains by calling the oracle with an arbitrary provisional set T_0 and using only the message component, which is independent of T_0 . No further loss is incurred.

Finally, a forgery of $\mathcal{A}$ is a forgery of $\mathcal{A}'$ on the same message, because the winning condition does not refer to the network model. Collecting the terms gives the stated inequality, where the two statistical terms are those of the weak game and are unchanged, since no additional oracle reprogramming was introduced. $\square$

C Proof of Theorem 4.4

PROOF OF THEOREM 4.4. Assume that the masking term of signer i is $\mathbf{m}_i = \Phi_i(\text{sd}_{\{i,j\}})_{j \in S_i, \text{pub}}$ for a set $S_i \subseteq T \setminus \{i\}$ and public data pub .

Necessity of a complete key graph. Suppose $h' \notin S_h$ for some pair $h \neq h'$ in T . Let the adversary corrupt $C = T \setminus \{h, h'\}$, which has size $t - 2$ and is admissible. Every seed occurring in $\mathbf{m}_h$ is of the form $\text{sd}_{\{h,j\}}$ with $j \in S_h \subseteq T \setminus \{h, h'\} = C$, so it is held by a corrupt party and the adversary computes $\mathbf{m}_h$. The adversary then runs

the attack of Theorem 3.3 with the roles unchanged, except that instead of replaying the last round it opens two sessions with signer sets T and T'' that differ in one index outside $\{h\}$; the Lagrange coefficients satisfy $\lambda_{T,h} \neq \lambda_{T'',h}$, and subtracting the two unmasked responses yields $(c\lambda_{T,h} - c''\lambda_{T'',h})s_h$ plus a difference of ephemeral terms whose distribution is known. Repeating this $\ell\varphi$ times gives a full-rank linear system in s_h over $\mathcal{R}_q$, exactly as in Appendix A, and recovers s_h . Hence $S_h = T \setminus \{h\}$ for every h .

Entropy of each pairwise term. Fix h and h' and let the adversary corrupt $C = T \setminus \{h, h'\}$. Write $\mathbf{m}_h = \Psi(\text{sd}_{\{h, h'\}}, \mathbf{k})$, where $\mathbf{k}$ collects the seeds shared with C , all of which the adversary knows. Conditioned on $\mathbf{k}$, the only unknown input of Ψ is $\text{sd}_{\{h, h'\}}$, so the conditional distribution P of $\mathbf{m}_h$ is a fixed distribution over $\mathcal{R}_q^\ell$ that the adversary can sample from. Suppose P is distinguishable from uniform with advantage ϵ . Since $\mathbf{z}_h = c\lambda_{T,h}s_h + f_h + \mathbf{m}_h$ and the adversary knows c , $\lambda_{T,h}$ and the distribution of f_h , the same distinguisher applied to $\mathbf{z}_h - x$ for a candidate value x of $c\lambda_{T,h}s_h$ separates the correct candidate from a wrong one with advantage $\epsilon - \text{negl}$, because for the correct candidate the residue is distributed as $f_h + \mathbf{m}_h$ and for a wrong one it is shifted by a nonzero element. Running the test over the coordinates of s_h one number theoretic transform slot at a time, the adversary recovers $\lambda_{T,h}s_h$ in φ slots with $O(\varphi\epsilon^{-2})$ sessions, hence s_h . Therefore ϵ must be negligible, that is the term keyed by $\text{sd}_{\{h, h'\}}$ must be indistinguishable from uniform over $\mathcal{R}_q^\ell$, and in particular an efficient generator of it must expand at least $\ell\varphi \log_2 q$ bits.

Independence over the choice of h' . The seeds are fixed at key generation, before the adversary selects the corruption set, so the argument above applies simultaneously to every $h' \in T \setminus \{h\}$. If two of the pairwise terms were correlated, say the ones keyed by $\text{sd}_{\{h, h'_1\}}$ and $\text{sd}_{\{h, h'_2\}}$, then an adversary corrupting $T \setminus \{h, h'_1\}$ would learn a function of the second term through the first, contradicting the previous paragraph. Therefore signer h must produce $t - 1$ mutually independent values, each indistinguishable from uniform over $\mathcal{R}_q^\ell$, which requires at least $(t - 1)\ell\varphi \log_2 q$ bits of pseudorandom output. $\square$

D Proof of Theorem 5.4

PROOF OF THEOREM 5.4. By Lemma 5.3 we have $\mathbf{W} = (\mathbf{A}\mathbf{z} - 2^{\text{ht}}\mathbf{c}) + \tilde{\mathbf{e}}$ with $\tilde{\mathbf{e}} = \sum_{j \in T} \mathbf{e}'_j + \boldsymbol{\delta} + \mathbf{c}\mathbf{e} - \mathbf{c}\mathbf{p}$. The combiner sets $\mathbf{h} = \lfloor \mathbf{W} \rfloor_{\text{w}} - \lfloor \mathbf{W} - \tilde{\mathbf{e}} \rfloor_{\text{w}}$, so the first component of the verification test holds identically. For the second, the rounding lemma of [6, Lemma 2.3] gives, for all $x, \delta \in \mathbb{Z}_q^m$,

$$\|2^v(\lfloor x + \delta \rfloor_v - \lfloor x \rfloor_v)\| \leq \|2^v \lfloor \delta \rfloor_v\| + 2^v \|\mathbf{1}\|,$$

whence $\|2^{v_w} \mathbf{h}\|_2 \leq \|\tilde{\mathbf{e}}\|_2 + 2^{v_w+1} \sqrt{k\varphi}$, and (4) follows from $\mathbf{z} = \mathbf{c}\mathbf{s} + \sum_j \mathbf{r}_j$ together with $\|\mathbf{c}\mathbf{s}\|_2 \leq \omega\sigma_t \sqrt{\ell\varphi} \ll \sigma_w \sqrt{t\ell\varphi}$.

It remains to bound the four summands of $\tilde{\mathbf{e}}$. The dominant one is $\sum_{j \in T} \mathbf{e}'_j$, a discrete Gaussian of parameter $\sigma_w \sqrt{t}$ in dimension $k\varphi$. The term $\boldsymbol{\delta}$ is a sum of t independent variables, each the rounding error of one coefficient and therefore uniform on an interval of length 2^{v_c} ; hence its coefficients have standard deviation $2^{v_c} \sqrt{t/12}$ and the ratio to the dominant term is $2^{v_c} / (\sigma_w \sqrt{12})$, independent of t . With $v_c = \lfloor \log_2 \sigma_w \rfloor - 1$ this ratio is $1/(2\sqrt{12}) = 0.144$, and since the two are independent the combined standard deviation is $\sqrt{1 + 0.144^2} = 1.0104$ times the dominant one, which is the claimed one per cent. The term $\mathbf{c}\mathbf{e}$ has coefficients of magnitude at most $\omega\sigma_t \sqrt{\varphi}$, which is $2^{29.7}$ at level I and therefore negligible against 2^{42} . The term $\mathbf{c}\mathbf{p}$ has coefficients with standard deviation at most

Table 10: Numerical verification of Theorem 5.4 at level I, over 200 trials per row.

t	$\max \ (z, 2^{v_w} \mathbf{h})\ _2$	B	ratio
4	$2.98 \cdot 10^{13}$	$4.52 \cdot 10^{14}$	0.066
16	$4.48 \cdot 10^{13}$	$4.52 \cdot 10^{14}$	0.099
64	$8.05 \cdot 10^{13}$	$4.52 \cdot 10^{14}$	0.178
256	$1.55 \cdot 10^{14}$	$4.52 \cdot 10^{14}$	0.344
1024	$3.10 \cdot 10^{14}$	$4.52 \cdot 10^{14}$	0.685

$\omega 2^{\text{ht}} / \sqrt{3}$, which is $2^{40.6}$ at level I; the ratio to the dominant term is 0.36, and this term is present in Threshold Raccoon as well, so it is already accounted for in the choice of B .

Finally, taking $B = \kappa \sigma_w \sqrt{T_{\max}(\ell + k)\varphi} + 2^{v_w+1} \sqrt{k\varphi}$ with κ the solution of $\kappa^2 - \log \kappa^2 = 1 + 2\lambda \log 2 / ((\ell + k)\varphi)$ makes the probability that a Gaussian of parameter $\sigma_w \sqrt{T_{\max}}$ in dimension $(\ell + k)\varphi$ exceeds the first summand at most $2^{-\lambda}$, by the standard tail bound. $\square$

As Table 10 shows, the second column is the largest Euclidean norm of the response together with the scaled hint over 200 Monte Carlo trials in which the aggregated randomness, the aggregated noise, the truncation perturbation $\boldsymbol{\delta}$ and the key rounding residue $\mathbf{c}\mathbf{p}$ are drawn from their exact distributions, and the third is the bound B of Theorem 5.4, a constant of the parameter set. The ratio stays below 0.69 at the largest supported threshold, so truncating to $v_c = 36$ bits leaves a margin of 1.46 and shortens the message of the second round from 15 680 to 4 160 bytes without weakening the bound.

E Proof of Theorem 5.5

PROOF OF THEOREM 5.5. *Notation.* Fix λ and a quantum adversary $\mathcal{A}$ making Q_s classical queries to (O_1, O_2, O_3) and Q_h superposition queries to (H_0, H_1, H_c) . Write $\mathcal{H} := [n] \setminus C$ for the honest indices, and for a session $s \in [Q_s]$ let $T^{(s)}$ be its signer set, $\mathcal{H}^{(s)} := \mathcal{H} \cap T^{(s)}$, and $\text{vw}_i^{(s)}$ the transcript delivered to $i \in \mathcal{H}^{(s)}$ by O_3 . Let $\sim_s$ be the equivalence on $\mathcal{H}^{(s)}$ given by $i \sim_s i' \iff \text{vw}_i^{(s)} = \text{vw}_{i'}^{(s)}$, let $\mathcal{P}^{(s)} := \mathcal{H}^{(s)} / \sim_s$ be its set of classes, and let

$$S^{\star(s)} := \begin{cases} \mathcal{H}^{(s)} & \text{if } |\mathcal{P}^{(s)}| = 1, \\ \perp & \text{otherwise,} \end{cases}$$

so that $S^{\star(s)}$ is the all-honest class when it exists. For G_r a game we write $\epsilon_r := \Pr[G_r = 1]$. Since the reduction holds $(\text{sk}_j)_{j \in C}$ and $|C| \leq t - 1$, for every T and every $h \in \mathcal{H} \cap T$ the identity

$$\lambda_{T,h} s_h = \mathbf{s} - \sum_{j \in T \setminus \{h\}} \lambda_{T,j} s_j \quad (5)$$

determines $\lambda_{T,h} s_h$ from $\mathbf{s}$ and the corrupt shares.

Game G_0 . The experiment $\text{Exp}_{\text{Thimza}, \mathcal{A}}^{\text{ts-uf-ca}}(1^\lambda)$ of Figure 3. By definition

$$\epsilon_0 = \text{Adv}_{\text{Thimza}, \mathcal{A}}^{\text{ts-uf-ca}}(\lambda). \quad (6)$$

Game G_1 . Replace every evaluation $\text{PRF}(\text{sd}_{\{i,j\}}, \cdot)$ with $\{i, j\} \subseteq \mathcal{H}$ by $F_{\{i,j\}}(\cdot)$, where the $F_{\{i,j\}}$ are independent uniformly random functions. A hybrid over the at most $\binom{|\mathcal{H}|}{2}$ honest seeds gives a distinguisher $\mathcal{B}_3$ with

$$|\epsilon_1 - \epsilon_0| \leq \text{Adv}_{\mathcal{B}_3}^{\text{PRF}}(\lambda). \quad (7)$$

The seeds are held by honest signers and queried only at points the honest code computes, so $\mathcal{B}_3$ makes classical queries and the step is valid against a quantum $\mathcal{A}$.

Game G_2 . Abort on the event

$$\text{Coll} := \exists H \in \{H_0, H_1, H_c\}, \exists x \neq x' : H(x) = H(x').$$

By the identical-until-bad lemma $|\varepsilon_2 - \varepsilon_1| \leq \Pr[\text{Coll}]$, and since the codomain has $2^{2\lambda}$ elements and $\mathcal{A}$ together with the honest oracles issues at most $Q_h + Q_s$ queries, the quantum collision bound of [36] gives an absolute constant C with

$$|\varepsilon_2 - \varepsilon_1| \leq \Pr[\text{Coll}] \leq C \frac{(Q_h + Q_s)^3}{2^{2\lambda}}. \quad (8)$$

Game G_3 . For each session s , at the moment the honest members of s publish their openings, reprogram H_c at the single point

$$x^{(s)} := (\text{vk}, \left[\sum_{j \in T^{(s)}} 2^{v_h} \bar{\mathbf{w}}_j \right]_{v_w}, \text{msg}^{(s)})$$

to a freshly sampled $c^{(s)} \leftarrow C$. Conditioned on the transcript before that moment, and using that G_2 excludes collisions of H_0 , the value $\bar{\mathbf{w}}_h$ of any $h \in \mathcal{H}^{(s)}$ is constrained only through $\text{cmt}_h = H_0(\text{vk}, \text{msg}, h, \bar{\mathbf{w}}_h)$, hence

$$H_{\infty}(x^{(s)} \mid \text{view}_{\mathcal{A}}) \geq \mu := k\varphi \log_2(q/2^{v_w}) - O(\lambda). \quad (9)$$

The Q_s reprogrammings therefore form an adaptive reprogramming experiment whose reprogrammed points each carry conditional min-entropy at least μ , and [37, Proposition 2] yields

$$|\varepsilon_3 - \varepsilon_2| \leq \frac{3Q_s}{2} \sqrt{Q_h 2^{-\mu}}. \quad (10)$$

No union bound over query points is taken, which is what makes the step sound against superposition access.

Game G_4 . For every session s and every class $S \in \mathcal{P}^{(s)}$ with $S \neq S^{\star(s)}$, replace $(z_i)_{i \in S}$ by independent uniform elements of $\mathcal{R}_q^\ell$. Fix such an S and pick $h \in S$ and $h' \in \mathcal{H}^{(s)} \setminus S$, which exists because $S \neq S^{\star(s)}$. The term

$$v := \varepsilon_{hh'} F_{\{h, h'\}}(v\mathbf{w}_h^{(s)})$$

occurs additively in $\mathbf{z}_h$; it occurs in no other response of the session, because $F_{\{h, h'\}}$ is held only by h and h' and h' evaluates it at $v\mathbf{w}_{h'}^{(s)} \neq v\mathbf{w}_h^{(s)}$. Hence v is uniform and independent of the rest of the transcript, so $\mathbf{z}_h$ is uniform and independent of $(z_i)_{i \neq h}$. Iterating over S gives that $(z_i)_{i \in S}$ is uniform, and therefore

$$\varepsilon_4 = \varepsilon_3. \quad (11)$$

This is Lemma 4.2 applied session by session.

Game G_5 . For each session with $S^{\star(s)} = \{h_1, \dots, h_a\} \neq \perp$, produce the responses by the simulator

$$\begin{aligned} \mathbf{z}_{h_u} &\leftarrow \mathcal{R}_q^\ell \quad (1 \leq u < a), \\ \mathbf{z}_{h_a} &:= \left(\text{cs} + \sum_{j \in T} \mathbf{r}_j \right) - \sum_{u < a} \mathbf{z}_{h_u} - \sum_{j \in T \setminus \mathcal{H}} (c\lambda_{T,j} \mathbf{s}_j + \mathbf{r}_j) + \Delta, \end{aligned} \quad (12)$$

where $\Delta := \sum_{u < a} \mathbf{m}_{h_u} + \mathbf{m}_{h_a}$ restricted to the terms keyed by seeds held by C , which the reduction computes from $(\text{sk}_j)_{j \in C}$. By Theorem 4.4 the masks of T sum to zero, so (12) is an identity rewriting of the honest responses of G_4 and

$$\varepsilon_5 = \varepsilon_4. \quad (13)$$

Game G_6 . Replace the pair

$$\left(\text{cs} + \sum_{j \in T} \mathbf{r}_j, \text{ce} + \sum_{j \in T} \mathbf{e}'_j \right)$$

of each session by an independent sample of $D_{\sigma_w \sqrt{\ell}}^\ell \times D_{\sigma_w \sqrt{\ell}}^k$. A distinguisher gives $\mathcal{B}_1$ against $\text{Hint-MLWE}_{q,k,\ell,Q_s,\sigma_1,\sigma_w,C}$ which receives $\mathbf{t}$ and hints $(c_u, c_u \mathbf{s} + \mathbf{r}_u, c_u \mathbf{e} + \mathbf{e}'_u)_{u \in [Q_s]}$ and answers the u -th signing query with the u -th hint:

$$|\varepsilon_6 - \varepsilon_5| \leq \text{Adv}_{\mathcal{B}_1}^{\text{Hint-MLWE}_{q,k,\ell,Q_s,\sigma_1,\sigma_w,C}}(\lambda). \quad (14)$$

The hint count is Q_s and not Q_h because c is fixed only at step 3 of O_3 , so the reduction knows which hint to consume when it answers a signing query and never when it answers a hash query.

Extraction. In G_6 the value s is used by no oracle, so $\mathcal{B}_2$ may embed a $\text{SelfTargetMSIS}_{q,k,\ell+1,C,\beta}$ instance in $[A \mid I_k]$ and publish $\mathbf{t}$. Let $(\text{msg}^\star, \sigma^\star = (c^\star, \mathbf{z}^\star, \mathbf{h}^\star))$ be a winning output, so that

$$c^\star = H_c(\text{vk}, [\mathbf{A}\mathbf{z}^\star - 2^{v_h} c^\star \mathbf{t}]_{v_w} + \mathbf{h}^\star, \text{msg}^\star), \quad (15)$$

$$\|(\mathbf{z}^\star, 2^{v_h} \mathbf{h}^\star)\|_2 \leq B, \quad \text{msg}^\star \notin \mathcal{M}. \quad (16)$$

Put $\mathbf{t}^\perp := \mathbf{A}\mathbf{s} + \mathbf{e} - 2^{v_h} \mathbf{t}$ for the key rounding residue and

$$\mathbf{y} := (\mathbf{z}^\star, 2^{v_h} \mathbf{h}^\star + \mathbf{t}^\perp c^\star, c^\star).$$

Unrolling the two roundings in (15) gives $H_c([A \mid I_k] \mathbf{y}, \text{msg}^\star) = c^\star$, and by (16) together with $\|c^\star\|_2 \leq \sqrt{\omega}$, $\|\mathbf{t}^\perp\|_\infty \leq 2^{v_h}$ and $\|\mathbf{h}^\star\|_\infty \leq 2$,

$$\|\mathbf{y}\|_2 \leq B + \sqrt{\omega} + (\omega 2^{v_h} + 2^{v_h+1}) \sqrt{k\varphi} = \beta, \quad (17)$$

so $\mathbf{y}$ is a $\text{SelfTargetMSIS}_{q,k,\ell+1,C,\beta}$ solution. It is nontrivial because $\text{msg}^\star \notin \mathcal{M}$, so no session reprogrammed H_c at a point carrying $\text{msg}^\star$. The extraction reads $\sigma^\star$ and performs no rewinding, hence applies verbatim to a quantum $\mathcal{A}$, and

$$\varepsilon_6 \leq \text{Adv}_{\mathcal{B}_2}^{\text{SelfTargetMSIS}_{q,k,\ell+1,C,\beta}}(\lambda). \quad (18)$$

Collecting. Chaining (6), (7), (8), (10), (11), (13), (14) and (18) through the triangle inequality $\varepsilon_0 \leq \varepsilon_6 + \sum_{r=0}^5 |\varepsilon_{r+1} - \varepsilon_r|$ yields

$$\begin{aligned} \text{Adv}_{\text{Thimza}, \mathcal{A}}^{\text{ts-uf-ca}} &\leq \text{Adv}_{\mathcal{B}_1}^{\text{Hint-MLWE}_{q,k,\ell,Q_s,\sigma_1,\sigma_w,C}} + \text{Adv}_{\mathcal{B}_2}^{\text{SelfTargetMSIS}_{q,k,\ell+1,C,\beta}} \\ &\quad + \text{Adv}_{\mathcal{B}_3}^{\text{PRF}} + C \frac{(Q_h + Q_s)^3}{2^{2\lambda}} + \frac{3Q_s}{2} \sqrt{\frac{Q_h}{2^\mu}}, \end{aligned}$$

which is the claim. $\square$

F Parameter derivation and its validation

Our derivation script implements the constraint system of [6, §3.1, §3.3] together with the standard gadget width of [39] and the correctness bound of Section 5.7, and estimates lattice security with the primal uSVP and primal SIS estimates under the core-SVP model [40, 41] with a classical cost of $2^{0.292\beta}$.

Two validations support it. First, on Kyber-512 the script returns a blocksize $\beta = 406$ and a classical core-SVP cost of 118.6 bits, and on the module learning with errors instance of Dilithium-2 it returns $\beta = 414$ and 120.9 bits, both within a fraction of a bit of the published figures. Second, instantiated with the structural choices of Ringtail at level I, namely $\varphi = 256$, $(\ell, k) = (7, 8)$, $\lceil \log_2 q \rceil = 48$ and gadget base 2^{12} , it reproduces their published parameter set, since it returns a challenge weight $\omega = 23$ against their 23, $\log_2 \sigma_u = 27.2$ against 27.2, $\log_2 \sigma_w = 37.35$ against 37.3, $\log_2 B = 48.51$ against 48.6, a preprocessing message of 602 112 bytes against 602 112 and an online message of 10 752 bytes against 10 752. Table 11 reports the design-space exploration this enables.

As Table 11 shows, each row is one structural choice for a scheme of the Ringtail type, with φ the ring degree, (ℓ, k) the module dimensions and $\bar{d} = 2k + k \lceil \log_b q \rceil$ the number of trapdoor columns

Table 11: Design space of the profile with two rounds, level I, $t = 1024$. Sizes in bytes.

φ	ℓ	k	$\bar{d}$	offline	online	$ \sigma $
256	7	8	48	602 112	10 752	14 208
1024	2	3	24	451 200	12 032	18 592
2048	1	1	10	129 536	11 776	16 160
2048	1	1	7	100 352	12 544	16 672

for gadget base b ; the first row is the published choice of Ringtail. A ring of higher degree with a module of rank one reduces $\bar{d}$ quadratically and shrinks the preprocessing by a factor of six, but it enlarges the smoothing term in the flooding constraint and raises the signature by six to seventeen per cent. No point dominates the first row in both columns, which is why we develop the profile with three rounds, where the commitment is not multiplied by a wide Gaussian vector and can be truncated instead.

G Strengthened game and the notion hierarchy

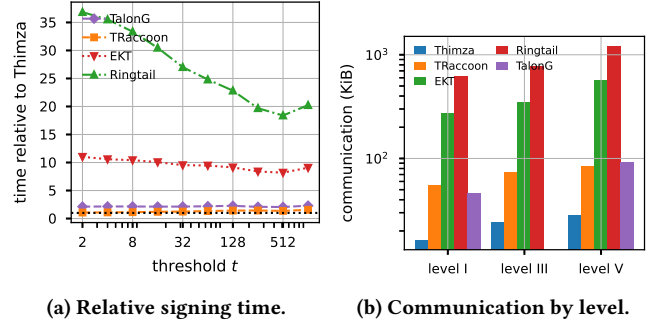
Bellare et al. [42] order the unforgeability notions for non-interactive threshold signatures from TS-UF-0, in which the forgery message must never have been submitted, to TS-UF-4, in which the adversary controls the signer sets and the accounting is per leader request. Our game of Definition 4.1 is orthogonal to that ladder, since it strengthens the *network* rather than the *bookkeeping*. The next statement addresses the transfer of the notion hierarchy to our model.

PROPOSITION G.1. *Let Σ be a canonical linear-response scheme. If Σ is TS-UF- x secure in the model with authenticated channels and first round aware of the signer set, and if the masking of Σ is view-derived with one seed per unordered pair, then Σ is TS-UF- x secure in the model of Definition 4.1, for every $x \in \{0, 1, 2, 3, 4\}$, with the same advantage up to an additive Adv^{PRF} .*

PROOF. The reduction of Appendix B does not inspect the winning predicate; it only translates oracle queries. Concretely it turns a query in the strong model into either a self-answered uniform value, when the querying signer lies in a view class other than the all-honest class, or into a query of the weak model, when it lies in the all-honest class. The set $\mathcal{M}$ of signed messages, or the per-leader counters of the higher notions, are updated identically in both games, because a query that the reduction answers itself corresponds to a class whose responses are uniform and therefore cannot help the adversary complete a signature, and a query that it forwards is recorded by the weak game. Hence the winning predicate is preserved and the advantage is unchanged up to the pseudorandomness term introduced in Hybrid 1. $\square$

Thimza satisfies TS-UF-0 in the weak model by Theorem 5.5, hence in the strong model by Proposition G.1. Whether it satisfies the higher notions is inherited from the underlying accounting and is orthogonal to the present contributions. The following statement addresses the min-entropy of the rounded aggregate commitment.

PROPOSITION G.2. *Let $\bar{\mathbf{w}}_h$ be the truncated commitment of an honest signer and let $\mathbf{W} = \sum_{j \in T} 2^{v_c} \bar{\mathbf{w}}_j$. Conditioned on the transcript of the first round and on the adversary's own choices, the min-entropy of $\lfloor \mathbf{W} \rfloor_{v_w}$ is at least $k\varphi \log_2(q/2^{v_w}) - \log_2(1/\epsilon_0)$, where ϵ_0 bounds the statistical distance between $\mathbf{A}\mathbf{r}_h + \mathbf{e}'_h$ and the uniform distribution over $\mathcal{R}_q^k$.*

**Figure 10: Relative speed and the effect of the security level.**

PROOF. Write $\mathbf{W} = 2^{v_c} \bar{\mathbf{w}}_h + \mathbf{U}$ where $\mathbf{U}$ collects the contributions of the other signers and is fixed once the first round closes. The value $\bar{\mathbf{w}}_h = \lfloor \mathbf{A}\mathbf{r}_h + \mathbf{e}'_h \rfloor_{v_c}$ is, before its opening, determined only through the hash cmt_h , and $\mathbf{A}\mathbf{r}_h + \mathbf{e}'_h$ is within statistical distance ϵ_0 of uniform over $\mathcal{R}_q^k$ because $\mathbf{r}_h$ has width $\sigma_w \geq \eta_\epsilon(\mathbb{Z}^\varphi)$ and $\mathbf{A}$ is uniform, by the regularity of the map $\mathbf{r} \mapsto \mathbf{A}\mathbf{r}$ over $\mathcal{R}_q$. Hence $\bar{\mathbf{w}}_h$ is within ϵ_0 of uniform over $\mathcal{R}_q^k$ truncated to $\log_2(q/2^{v_c})$ bits per coefficient, which is 13 bits at the level I parameters. Rounding the sum to v_w bits maps exactly $2^{v_w - v_c}$ values of each coefficient of $\bar{\mathbf{w}}_h$ to the same output, so the output is again within ϵ_0 of uniform on its own range, whose logarithm is $k\varphi \log_2(q/2^{v_w})$; the min-entropy of a distribution within statistical distance ϵ_0 of the uniform distribution on a set of size N is at least $\log_2 N - \log_2(1/\epsilon_0)$. $\square$

At the level I parameters, $k\varphi \log_2(q/2^{v_w}) = 2560 \cdot 9 = 23040$ bits, so μ is at least that minus a negligible correction, so that the reprogramming term $\frac{3Q_s}{2} \sqrt{Q_h} 2^{-\mu}$ of Game G_3 of Appendix E stays below $2^{-\lambda}$ with $\lambda = 128$ by a wide margin, the square-root loss halving the available entropy and leaving more than 11 000 bits of slack.

As Figure 10a presents, the measured signing time of each prior scheme divided by that of Thimza stays above one over the whole range, and each curve has a shape that identifies its cause. The ratio to Ringtail falls from 37 to 20, because the fixed cost of its wide preprocessing dominates at small t and the linear term catches up. The ratio to Threshold Raccoon rises from 1.07 to 1.56, since the two schemes do almost the same work until the masking starts to grow. The ratios to Hermine, to the scheme of [5] and to TalonG are nearly flat, at about 6, 10 and 2, because in those the two effects cancel.

As Figure 10b presents, communication per signer is grouped by security level on a logarithmic axis, and a missing bar means that the scheme has no published parameter set at that level. The bars of the schemes that preprocess grow faster from level to level, because the width of their first object is proportional to the module dimension times the gadget length and both grow. The bar of Thimza grows slowest of the interchangeable schemes, since its transcript is one truncated commitment plus one response and neither carries a trapdoor factor. At level V the gap to Ringtail is a factor 43.

H Adapting the construction to two rounds

It is natural to ask whether the techniques of Section 4 can be applied to a profile with two rounds, and the answer is that they can, but that the bandwidth advantage does not survive. We record the analysis because it delimits the contribution.

Table 12: Validation of the estimator against published parameter sets.

Instance	N	$\lceil \log_2 q \rceil$	our β / bits	published bits
Kyber-512	512	12	406 / 118.6	118
Kyber-768	768	12	624 / 182.2	183
Kyber-1024	1024	12	874 / 255.2	256
Dilithium-2 (MLWE)	1024	23	414 / 120.9	123
Dilithium-3 (MLWE)	1280	23	519 / 151.5	—

The view-partition lemma and the antisymmetric masking are independent of the round structure and apply verbatim, so a scheme with two rounds built on the Ringtail blueprint with view-derived antisymmetric masks needs neither authenticated channels nor a first round aware of the signer set, and needs only session-bounded state. What does not carry over is the truncation of the commitment. In a scheme with three rounds the aggregate is $\sum_j \mathbf{w}_j$ and the truncation perturbs it by a quantity of standard deviation $2^{v_c} \sqrt{t/12}$, which is compared against the natural noise $\sigma_w \sqrt{t}$; the ratio does not depend on t and permits $v_c = \lfloor \log_2 \sigma_w \rfloor - 1$. In a scheme with two rounds the aggregate is $\sum_j \mathbf{W}_j \mathbf{u}$ for an aggregation vector $\mathbf{u}$ of width σ_u , so the perturbation becomes $2^{v_c} \sigma_u \sqrt{t d \phi / 12}$ and the admissible truncation shrinks to

$$v_c \leq \log_2 \left(\frac{\sigma_w}{\sigma_u} \right) - \frac{1}{2} \log_2 (\bar{d} \phi) - 2,$$

which at the Ringtail parameters is $v_c \leq 3$ and therefore saves six per cent rather than a factor 3.8. The second obstacle is the trapdoor width itself, since the object of the first round has $k(\bar{d} + 1)$ ring entries with $\bar{d} = 2k + k \lceil \log_b q \rceil$, so it is quadratic in the module dimension, and Table 11 shows that reducing it by moving to a ring of higher degree costs six to seventeen per cent in signature size. This is why we develop the profile with three rounds, and why the future direction stated in Section 7 is the transfer of the query accounting rather than of the truncation.

I Security estimation methodology

All bit-security numbers in this paper are produced by one script applied uniformly to every scheme, so that no difference between schemes is an artefact of a different estimation convention. We use the core-SVP methodology [40, 41]. For a blocksize β the root Hermite factor is

$$\delta_\beta = \left(\frac{(\pi\beta)^{1/\beta} \beta}{2\pi e} \right)^{\frac{1}{2(\beta-1)}},$$

and the classical cost of one call to a lattice reduction with that blocksize is $2^{0.292\beta}$, the quantum cost $2^{0.257\beta}$. For a module learning with errors instance of secret dimension $N = \ell\phi$, sample dimension $M = k\phi$, modulus q and width σ we take the smallest β for which the primal uSVP condition

$$\sqrt{\beta} \sigma \leq \delta_\beta^{2\beta-d-1} q^{(d-N-1)/d}$$

holds for some embedding dimension $d \leq N + M + 1$. For a short integer solution instance with N equations, M unknowns, modulus q and Euclidean bound B we take the smallest β for which $\delta_\beta^d q^{N/d} \leq B$ holds for some $d \leq M$, optimising d around $\sqrt{N \log q / \log \delta_\beta}$.

When a public value is published in rounded form we replace σ by $\max(\sigma, 2^v / \sqrt{12})$, since the rounding residue is the effective noise.

Table 13: Communication per signer, signing time by round, in milliseconds. Key generation is the whole dealer cost for n key holders.

Scheme	t	KeyGen	round 1	round 2	round 3	signing
Thimza	2	1.81	1.05	0.00	1.04	2.09
	8	3.30	1.03	0.00	1.51	2.54
	32	10.55	1.15	0.00	3.32	4.47
	64	22.66	1.06	0.00	5.76	6.82
	128	56.02	1.14	0.00	10.37	11.51
	256	182.35	1.10	0.00	21.71	22.81
	1024	2258.55	1.19	0.00	77.10	78.29
TRaccoon	2	2.35	1.13	0.00	1.10	2.23
	8	3.39	1.31	0.01	1.56	2.89
	32	15.23	2.16	0.04	3.36	5.56
	64	42.00	3.27	0.07	5.74	9.07
	128	135.63	5.57	0.14	10.61	16.32
	256	506.16	10.63	0.29	21.73	32.64
	1024	7526.90	40.23	1.20	80.57	122.00
Hermine	2	1.18	6.07	10.73	—	16.81
	4	1.29	5.76	12.10	—	17.87
	8	1.43	5.57	13.63	—	19.20
	16	1.71	5.71	16.15	—	21.85
	32	2.58	5.50	22.78	—	28.28
	64	3.93	6.14	34.82	—	40.96
TalonG	2	2.40	2.20	2.24	—	4.44
	8	4.93	2.20	3.27	—	5.47
	32	18.32	2.22	7.31	—	9.53
	64	41.98	2.21	12.77	—	14.97
	128	109.36	2.32	23.79	—	26.10
	256	326.07	2.20	45.36	—	47.57
	1024	4058.67	2.32	176.88	—	179.20
EKT	2	3.07	10.31	12.71	—	23.02
	8	4.44	10.39	16.12	—	26.51
	32	13.45	10.36	32.23	—	42.59
	64	30.95	10.35	54.13	—	64.47
	128	85.10	10.22	94.81	—	105.03
	256	277.31	10.34	181.20	—	191.54
	1024	3793.67	10.79	699.21	—	710.00
Ringtail	2	2.87	24.64	52.34	—	76.99
	8	3.82	23.81	61.08	—	84.89
	32	15.00	24.08	96.70	—	120.78
	64	41.77	25.68	143.43	—	169.12
	128	135.54	25.29	237.47	—	262.76
	256	490.41	23.91	425.19	—	449.10
	1024	7308.54	25.43	1559.00	—	1584.43

As Table 12 shows, each row is a standardised parameter set whose security is public knowledge, N is the secret dimension in scalar terms, the third column is the blocksize our script returns together with the classical cost 0.292β , and the last is the figure reported by the designers. The agreement is within three bits on every row and exact on the blocksize for Kyber-512, which is the usual sanity check and what licences the use of the same script for Table 11 and for the re-derivation reported in Appendix F.

J Extended measurements

Table 13 gives the complete benchmark, one row per scheme and threshold, with the cost of every round separated. All entries are the minimum of three executions on a single core, and the key generation column is the cost borne once by the dealer for $n = t$ key holders, which is why it grows superlinearly.

As Table 13 shows, rows are grouped by scheme and ordered by threshold, and a dash marks a round that does not exist. Four regularities appear. The first round of Thimza stays at 1.1 milliseconds because it depends on neither t nor a row mask, whereas that of Threshold Raccoon grows from 1.13 to 40.23 because its row mask costs t pseudorandom expansions. The second round of Thimza is free, since it forwards a stored value. The third rounds of the two schemes grow linearly with slopes in the ratio of Corollary 4.5. The first round of every scheme that preprocesses is constant but large, at 5.7, 10.3 and 25 milliseconds for Hermine, the scheme of [5]

and Ringtail, which is the price of a wide object, and their online round absorbs everything else. Key generation is a one time cost dominated by the sharing, and it is the one column where Hermine is ahead, since its recursive sharing at $n = t$ produces a single share per party.

K Remarks on the lower bounds

The two lower bounds of Section 4 are stated for the class of Definition 3.1, and it is worth recording what happens outside it, because this delimits where a future improvement could come from.

Schemes with short reconstruction coefficients. If the secret sharing has reconstruction coefficients of small norm, then $cy_i s_i$ is short and the ephemeral randomness floods it, so no mask is needed and both Theorem 4.4 and Proposition 4.6 become vacuous; the message of the last round is then a short vector and can be encoded in $\ell\varphi(\log_2 \sigma_w + O(1))$ bits rather than $\ell\varphi \log_2 q$, which at the level I parameters would be 11 418 instead of 12 544 bytes. This is the route taken by [7], and it is expensive for a different reason, namely that the only known sharings with short coefficients over $\mathcal{R}_q$ have shares that grow polynomially in n ; for the parameters of [7] at $n = 5$ the share growth costs a signature of 258 KiB. A sharing over $\mathcal{R}_q$ with both short coefficients and compact shares would improve every scheme in the family at once, and its existence is open.

Interactive masking. Theorem 4.4 assumes that the mask is computed non-interactively from pairwise keys. If the signers are allowed one extra round to agree on a mask, for instance by running a small multiparty protocol, the pairwise structure is no longer forced. The bound then does not apply, but the extra round costs at least one round trip, which by the model of Section 7 is 170 milliseconds in a wide area deployment against the 80 milliseconds that the whole masking currently costs, so this trade is not favourable at the thresholds considered here.

Correlated randomness from a setup. If the dealer distributes, at key generation, a large table of precomputed zero-sum masks, then the online cost drops to a table lookup. The bound is not contradicted, since the pseudorandom material is still generated, only earlier; the cost reappears as storage of $\Theta(Q_s t \ell\varphi \log_2 q)$ bits per signer, which for $Q_s = 2^{60}$ is not a deployment option. The pseudorandom function is exactly the standard way of compressing that table, and Theorem 4.4 says that the compression cannot also reduce the online work.

Relation to the impossibility of statelessness. Theorem 3.3 and Theorem 4.4 have the same source, namely that an adversary controlling $t - 1$ or $t - 2$ parties can strip every quantity that is a deterministic function of keys it holds. The first uses this to remove the mask entirely and replay, the second to isolate a single pairwise term. It follows that any construction that evades one of them by changing the masking mechanism must be checked against the other.

L Encoding conventions

All byte counts in this paper are produced by a single serializer with the following rules, so that they are comparable across schemes. A uniformly distributed element of $\mathcal{R}_q$ occupies $\varphi[\log_2 q]$ bits, since it is incompressible. An element whose coefficients are Gaussian with parameter σ occupies $\varphi[\log_2 \sigma + 2.6]$ bits, which is the entropy of a discrete Gaussian rounded up to the nearest bit plus the half bit lost by a practical range coder. A value in $\mathcal{R}_q$ published with ν least significant bits dropped occupies $\varphi[\log_2(q/2^\nu)]$ bits. A hash output or a seed occupies 2λ bits. The verification key of every

Raccoon-type scheme consists of a seed for A and the rounded key, and the signature consists of a challenge seed, the response and the hint.

These rules reproduce the published figures closely. For Threshold Raccoon they give a verification key of 3 872 bytes against a published 3 856, a signature of 13 152 bytes against a published upper bound of 12 736, and a transcript per signer of $40\,848 + 16t$ bytes against a published $40\,800 + 16t$. For Ringtail they give a preprocessing message of 602 112 bytes and an online message of 10 752 bytes, both equal to the published values. The signature figures are conservative by three to four per cent because the published encoders use an arithmetic coder rather than a fixed-width one; since the same rule is applied to every scheme, including ours, the comparison is not affected.

M Attack on masking derived from an identifier

We go through the attack sketched in Section 3, which is what forces the registry of [4] to be a set. Let Σ use masks $\mathbf{m}_i(\text{sid})$ that depend only on a session identifier, and let a signer accept an identifier at most once per session but not across sessions with distinct tokens.

Algorithm 4 Key-share recovery from masks derived from an identifier

- 1: corrupt C with $|C| = t - 1$, let $h \notin C$ and fix sid
 - 2: **for** $u = 1$ to $\ell\varphi$ **do**
 - 3: pick $T_u \ni h$ with $|T_u| = t$ and $T_u \setminus \{h\} \subseteq C \cup \{j_u\}$
 - 4: run a full session with signer set T_u , identifier sid and a fresh message msg_u , obtaining $z_h^{(u)}$
 - 5: compute $\mathbf{m}_h(\text{sid})$ from the corrupt seeds and set $\text{Hi}_u := z_h^{(u)} - \mathbf{m}_h(\text{sid})$
 - 6: **end for**
 - 7: solve $\{\text{Hi}_u = c_u \lambda_{T_u, h} \mathbf{s}_h + \mathbf{r}_h^{(u)}\}_u$ for $\mathbf{s}_h$ by lattice reduction on the q -ary lattice spanned by the coefficients $c_u \lambda_{T_u, h}$
 - 8: **return** $\mathbf{s}_h$, hence s by interpolation with the corrupt shares
-

As Algorithm 4 presents, the adversary first fixes a corruption set of maximal size and an identifier it will reuse, with the honest index h as its target. It then opens $\ell\varphi$ sessions that share the identifier but differ in the signer set, which a signer guarding only against repetition inside one session will accept. Since h is the only honest member of every set, every seed it uses is shared with a corrupt party, so the adversary recomputes the mask and strips it. What remains is $\ell\varphi$ noisy linear equations in $\mathbf{s}_h$ whose coefficients are uniform in $\mathcal{R}_q$ and adversarially chosen and whose noise has width $\sigma_w = 2^{37}$ against a modulus 2^{49} ; the resulting lattice has a unique short vector far below the Gaussian heuristic, so reduction succeeds and interpolation with the corrupt shares finishes the recovery. This is why the mask must be bound to something the adversary cannot repeat while changing the Lagrange coefficients, and the view is the smallest such object.

N Supporting subroutines

For completeness we fix the auxiliary functions used by Algorithms 1 to 3. All of them are instantiated from one extendable output function, which we take to be SHAKE256, with domain separation by a one-byte tag.

Algorithm 5 Subroutines of *Thimza*

```

Expand(seed)  $\rightarrow \mathbf{A} \in \mathcal{R}_q^{k \times \ell}$ 
1: absorb ( $0 \times 01$ , seed), squeeze  $k\ell\varphi$  words of 64 bits, reduce each
   modulo  $q$ 
   SampleG(seed,  $d, \sigma$ )  $\rightarrow D_\sigma^d$ 
2: absorb ( $0 \times 02$ , seed,  $\sigma$ ), squeeze  $d\varphi$  words, map each through
   the inverse Gaussian cumulative distribution and round
    $H_0, H_1$ 
3: absorb ( $0 \times 03$  resp.  $0 \times 04$ , length-prefixed inputs), squeeze  $2\lambda$ 
   bits
    $H_c(\mathbf{vk}, \mathbf{w}, \text{msg}) \rightarrow C$ 
4: absorb ( $0 \times 05$ ,  $\mathbf{vk}, \mathbf{w}, \text{msg}$ ), squeeze; then draw  $\omega$  distinct posi-
   tions in  $[\varphi]$  and one sign each, rejecting repeated positions
    $\text{PRF}(\text{sd}, \mathbf{vw}) \rightarrow \mathcal{R}_q^\ell$ 
5: absorb ( $0 \times 06$ ,  $\text{sd}, \mathbf{vw}$ ), squeeze  $\ell\varphi$  words of 64 bits, reduce each
   modulo  $q$ 

```

As Algorithm 5 presents, each block is one subroutine, given by the string it absorbs and by the way the squeezed bits become the required object. The one byte tags are domain separators that keep the six functions independent although they share one primitive, which is what lets the proof treat them as independent quantum-accessible random oracles. The rejection inside the challenge sampler makes its output uniform on the set C of ternary polynomials of weight ω , and reducing 64 bit words modulo q costs a bias of at most 2^{-15} per coefficient at these moduli, absorbed by the statistical terms of Theorem 5.5. The Gaussian sampler is written through the inverse cumulative distribution because both widths in use are far above the smoothing parameter, so a table or a rounded continuous sampler are within negligible distance of the discrete Gaussian.

O Standardization considerations

We state how *Thimza* relates to the criteria of the ongoing call for threshold schemes [3].

Category and profile. *Thimza* is a signing scheme in the category of schemes built on a primitive that is not itself a published standard, since its counterpart with one signer is the Raccoon signature. Its threshold profile is (t, n) for any $1 \leq t \leq n \leq 1024$, with a dishonest majority and static corruption.

Interoperability. By Theorem 5.1 a *Thimza* signature is a Raccoon signature. A relying party therefore needs no change, which is the functional interchangeability that the call singles out as a criterion. *Thimza* is not functionally *equivalent* to Raccoon, since the distribution of the aggregated randomness is a sum of t Gaussians rather than one, and the hint carries the truncation perturbation; Theorem 5.1 bounds the resulting statistical distance.

Setup. Key generation is performed by a trusted dealer, as in [4–6, 8]. A distributed key generation can replace it, and any protocol that outputs a Shamir sharing of a short s over $\mathcal{R}_q$ together with pairwise seeds is a drop-in substitute, since Theorem 5.5 only uses that the shares of the corrupt parties are uniform and that the seeds shared by two honest parties are unknown.

Network. By Theorem 4.3 unforgeability needs no authenticated channels. Liveness does need them, in the sense that an adversary controlling the network can always make an execution fail; this is the standard separation between unforgeability and robustness and it is shared by every scheme in Table 1.

State. By Corollary 3.6 a signer holds one ephemeral token per open session and no persistent counter or registry. This matters

Table 14: Modelled $\mathcal{R}_q$ -operations per signer, in units of 10^3 , at $t = 1024$.

Scheme	level I	level III	level V
Thimza	4 904	7 270	8 464
TRaccoon	7 034	10 424	12 137
TalonG	10 577	—	21 154
EKT	51 302	83 617	109 650
Ringtail	113 705	159 187	207 850

for the call, which treats state as a deployment hazard, and it is the property that distinguishes *Thimza* from [4].

Side channels. Every operation of Algorithm 2 other than the hash calls is linear over $\mathcal{R}_q$, so the additive sharing used by masked implementations of Raccoon applies unchanged, and the number of shares is a local choice of each signer.

P Complexity at all levels

Table 14 evaluates the model of Table 4 at the three levels, which shows how the balance between the commitment and the masking term moves with the parameters.

As Table 14 shows, each entry totals the three columns of Table 5 at the parameter set of that level and $t = 1024$; *Hermine* is absent because it has no set at that threshold. All schemes grow by a factor between 1.7 and 2.1 from level I to level V, which is the growth of $\ell\varphi$ and of the module dimension, and the ordering never changes. The ratio of *Thimza* to Threshold Raccoon stays near 1.4, the constant predicted by the masking coefficient, and the ratio to Ringtail stays above 20 because the trapdoor width grows faster than the masking term.

Q The simulator in pseudocode

For concreteness we give the simulator used in Game G_5 of Appendix E. It receives the verification key and the Q_s hints of the Hint-MLWE instance, it holds the corrupt shares, and it never uses $\mathbf{s}$.

Algorithm 6 Simulation of a signing session

```

On  $O_1(i, \text{msg})$  for  $i \in \mathcal{H}$ 
1:  $\bar{\mathbf{w}}_i \leftarrow \mathcal{R}_q^k$  truncated to  $\lceil \log_2 q \rceil - v_c$  bits per coefficient
2: store the token  $(\text{eid}, i, \bar{\mathbf{w}}_i)$ ; return  $H_0(\mathbf{vk}, \text{msg}, i, \bar{\mathbf{w}}_i)$ 
On  $O_3(\text{eid}, i, T, (\text{cmt}_j, \bar{\mathbf{w}}_j)_{j \in T})$ 
3: compute  $\mathbf{vw}_i$ ; group the pending honest queries of this session
   by  $\mathbf{vw}$ 
4: if the class of  $i$  is not the all-honest class then
5:   return  $\mathbf{z}_i \leftarrow \mathcal{R}_q^\ell$ 
6: end if
7: if  $i$  is not the last honest signer of its class to be queried then
8:   return  $\mathbf{z}_i \leftarrow \mathcal{R}_q^\ell$ 
9: end if
10:  $\mathbf{W} := \sum_{j \in T} 2^{v_c} \bar{\mathbf{w}}_j$ ;  $\mathbf{w} := \lfloor \mathbf{W} \rfloor_{v_w}$ 
11: if  $H_c$  is defined at  $(\mathbf{vk}, \mathbf{w}, \text{msg})$  then abort
12: consume the next hint  $(c, \text{Hi} = \mathbf{cs} + \mathbf{r}, \text{Hi}' = \mathbf{ce} + \mathbf{e}')$ 
13: program  $H_c(\mathbf{vk}, \mathbf{w}, \text{msg}) := c$ 
14:  $\mathbf{z}_i := \text{Hi} - \sum_{u < a} \mathbf{z}_{h_u} - \sum_{j \in T \setminus \mathcal{H}} (c\lambda_{T,j} \mathbf{s}_j + \mathbf{r}_j) + \Delta$ 
15: return  $\mathbf{z}_i$ 

```

Table 15: Transcript components in bytes at level I and $t = 1024$.

Scheme	component	bytes	formula
Thimza	commitment hash	32	$2\lambda/8$
	truncated opening	4 160	$k\varphi(L - v_c)/8$
	masked response	12 544	$\ell\varphi L/8$
TRaccoon	commitment hash	32	$2\lambda/8$
	row mask	12 544	$\ell\varphi L/8$
	opening + tags	15 680 + 16 384	$k\varphi L/8 + 16t$
	masked response	12 544	$\ell\varphi L/8$
EKT	preprocessing matrix	264 000	$k\rho\varphi L/8$
	masked response	14 400	$\ell\varphi L/8$
Ringtail	preprocessing matrix	602 112	$k(\bar{d} + 1)\varphi L/8$
	masked response	10 752	$\ell\varphi L/8$
	tags	16 384	$16t$
TalonG	semi-commitment	11 008	$\varphi L/8$
	masked $\mathbf{z}, \Delta$	22 016	$2\varphi L/8$
	opening r_1, r_2	13 824	$2\varphi[\log_2 \sigma_r + 2.6]/8$
Hermine	truncated commitments	53 248	$k \text{rep } \varphi(L - v_w)/8$
	short response	20 992	$(\ell + k)\varphi[\log_2 \sigma_z + 2.6]/8$

As Algorithm 6 presents, a query of the first round is answered without any secret by sampling the truncated commitment uniformly, which is correct because the real $\bar{\mathbf{w}}_i$ is within negligible distance of uniform by Proposition G.2, and by storing the token and returning its hash. For a query of the last round the simulator computes the view of the querying signer and the partition of the honest signers, then answers with a uniform value every signer whose response is uniform, which by Lemma 4.2 is every signer outside the class containing all honest members and, inside that class, every signer but the last one to be queried. That last signer anchors the session: the simulator rebuilds the aggregate, aborts if the challenge oracle was already fixed there, which is the event bounded by the min-entropy term of Game G_3 , draws the one hint of the Hint-MLWE instance that this session consumes, programs the challenge, and sets the response so that the sum over the session equals the value the hint provides, where Δ collects the mask terms keyed with corrupt seeds. Since a hint is drawn only here, the number of hints equals the number of signing queries and not the number of oracle queries, and the secret $\mathbf{s}$ is never touched, which is what makes the final extraction possible.

R Size accounting by component

Table 15 presents every transcript into its components, so that it can be checked each entry of Table 6 against the formulas of Table 4.

As Table 15 shows, each block is one scheme and each row one transmitted object with the formula that produces it, where $L = \lceil \log_2 q \rceil$ and $\rho, \bar{d}, \text{rep}$ are as in Section 6. The Thimza block shows that two of its three components are already minimal, the hash by the collision requirement and the response by Proposition 4.6. The Threshold Raccoon block shows the three items we remove or shrink, namely the row mask, the $16t$ bytes of tags and the untruncated opening. The blocks of the schemes that preprocess show that one component, the object of the first round, is 72, 94 and 96 per cent of the total for Hermine, the scheme of [5] and Ringtail. The TalonG block shows the two objects that make its signature the largest, and the opening is exactly what its verification equation needs and therefore what excludes it from interchangeability.

S Latency model observation

Table 16 lists the terms of the model of Section 7, so that it can be substituted a different network. As Table 16 shows, “on-rd” is the

Table 16: Modelled wide area latency in milliseconds, with a round trip time of 170 and a throughput of 183 megabits per second.

Scheme	t	on-rd	on bytes	transfer	compute	online / total
Hermine	64	1	20 992	0.9	40.96	211.9 / 384.2
EKT	64	1	14 400	0.6	64.47	235.1 / 416.6
Ringtail	64	1	11 776	0.5	169.12	339.6 / 535.9
TalonG	64	2	46 848	2.0	14.97	357.0 / 357.0
Thimza	64	3	16 736	0.7	6.82	517.5 / 517.5
TRaccoon	64	3	41 824	1.8	9.07	520.9 / 520.9
TalonG	1024	2	46 848	2.0	179.20	521.2 / 521.2
Thimza	1024	3	16 736	0.7	78.29	589.0 / 589.0
TRaccoon	1024	3	57 184	2.5	122.00	634.5 / 634.5
EKT	1024	1	14 400	0.6	710.00	880.6 / 1062.2
Ringtail	1024	1	27 136	1.2	1584.43	1755.6 / 1951.9

number of rounds that depend on the message and therefore the number of round trips a user waits for, “on bytes” is what a signer sends in them, “transfer” is those bytes over the throughput, “compute” is the measurement of Table 8, and the last column gives the latency of the part that depends on the message and of a complete execution. Three observations follow. The transfer term is under three milliseconds everywhere, so at this throughput bandwidth reaches latency only through the preprocessing of the schemes that have it. At $t = 64$ the round trips are everything and the ranking follows the round count. At $t = 1024$ the compute term takes over, which pushes the scheme of [5] and Ringtail behind despite their single online round and leaves Thimza and TalonG within thirteen per cent of each other. A faster network moves the ranking towards fewer rounds, and a slower one towards fewer bytes and less computation.

T A worked execution

We trace one execution at the level I parameters with $t = 3$ and $n = 5$, which is the smallest profile in our validation suite that exercises every mechanism, and we record the size and the purpose of each quantity. Let $\varphi = 512$, $\lceil \log_2 q \rceil = 49$, $\ell = 4$, $k = 5$, $\omega = 19$, $v_t = 37$, $v_w = 40$ and $v_c = 36$.

Setup. The dealer expands seed_A into $\mathbf{A} \in \mathcal{R}_q^{5 \times 4}$, samples $\mathbf{s}$ and $\mathbf{e}$ of width 2^{20} , publishes $\mathbf{t} = \lfloor \mathbf{A}\mathbf{s} + \mathbf{e} \rfloor_{37} \in \mathcal{R}_{q^{212}}^5$, which occupies $5 \cdot 512 \cdot 12/8 = 3840$ bytes, and shares $\mathbf{s}$ with a polynomial of degree 2 over $\mathcal{R}_q^4$. It draws $\binom{5}{2} = 10$ pairwise seeds, so signer 1 stores four of them, 128 bytes in total, against the 320 bytes that an ordered construction would need.

Round 1. Signer 1 samples $\mathbf{r}_1 \in \mathcal{R}_q^4$ and $\mathbf{e}'_1 \in \mathcal{R}_q^5$ of width 2^{37} and forms $\mathbf{w}_1 = \mathbf{A}\mathbf{r}_1 + \mathbf{e}'_1$, which costs twenty ring products in the transform domain. It truncates to $\bar{\mathbf{w}}_1 = \lfloor \mathbf{w}_1 \rfloor_{36}$, an element of $\mathcal{R}_q^5$ with 13 bits per coefficient, and publishes only $H_0(\text{vk}, \text{msg}, 1, \bar{\mathbf{w}}_1)$, which is 32 bytes. Nothing in this message names the signer set, so the same token could later be used in any coalition containing signer 1; what fixes the coalition is which signers answer in the next round.

Round 2. Signer 1 publishes $\bar{\mathbf{w}}_1$, which is $5 \cdot 512 \cdot 13/8 = 4160$ bytes. Threshold Raccoon publishes the untruncated $\mathbf{w}_1$ here, 15 680 bytes, together with a tag for each partner and, in its first round, a row mask of 12 544 bytes.

Round 3. The coalition is $T = \{1, 2, 3\}$. Signer 1 verifies the three openings against the three hashes, forms the digest vw of the whole transcript, rebuilds $\mathbf{W} = \sum_j 2^{36} \bar{\mathbf{w}}_j$, rounds it to $\mathbf{w} = \lfloor \mathbf{W} \rfloor_{40} \in \mathcal{R}_{q_{29}}^5$ and derives $c \in C$ with 19 nonzero ternary coefficients. Its mask is $\varepsilon_{12}\text{PRF}(\text{sd}_{\{1,2\}}, \text{vw}) + \varepsilon_{13}\text{PRF}(\text{sd}_{\{1,3\}}, \text{vw})$, two evaluations rather than six, with $\varepsilon_{12} = \varepsilon_{13} = +1$ since 1 is the smallest index. It outputs $\mathbf{z}_1 = c\lambda_{T,1}\mathbf{s}_1 + \mathbf{r}_1 + \mathbf{m}_1$, which is $4 \cdot 512 \cdot 49/8 = 12\,544$ bytes, and erases $(\mathbf{r}_1, \bar{\mathbf{w}}_1)$.

Combination. The three masks cancel, since $\varepsilon_{12} + \varepsilon_{21} = 0$ and likewise for the other two pairs, so $\mathbf{z} = c\mathbf{s} + \mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3$. The hint is $\mathbf{h} = \mathbf{w} - \lfloor \mathbf{A}\mathbf{z} - 2^{37}c\mathbf{t} \rfloor_{40}$ and its coefficients lie in $\{-2, \dots, 2\}$ at this threshold. The signature $(c, \mathbf{z}, \mathbf{h})$ is 13 152 bytes and is accepted by the Raccoon verifier.

Totals. Signer 1 sends $32 + 4\,160 + 12\,544 = 16\,736$ bytes and generates two pseudorandom vectors. In the same execution a Threshold Raccoon signer sends $32 + 12\,544 + 15\,680 + 48 + 12\,544 = 40\,848$ bytes and generates six.

U Relation to standard unforgeability notions

Definition 4.1 strengthens the network model of the unforgeability game but leaves its winning condition untouched, so it is worth mentioning how it sits against the other notions. The following statement addresses the position of our game in the hierarchy of [42].

PROPOSITION U.1. *The game of Definition 4.1 is at least as strong as TS-UF-0 of [42]: every adversary that wins the latter against a canonical linear-response scheme also wins the former, with the same advantage and the same running time.*

PROOF. The two games agree on key generation, on the corruption phase and on the winning condition, which in both is that the output is a valid signature on a message that was never submitted to a signing oracle. They differ only in the oracle interface, and every query admissible in TS-UF-0 is admissible in Definition 4.1, since the latter drops the argument T of the first oracle and drops the check that the messages of the other honest signers in the first round are genuine. An adversary against TS-UF-0 is therefore also an adversary against our game, and it is answered identically on every query it makes. $\square$

The next statement addresses the guarantee that a relying party obtains.

THEOREM U.2. *Let Σ' be the Raccoon signature scheme with the parameters of Table 3. For every quantum polynomial time adversary $\mathcal{A}$ with superposition access to H_0, H_1 and H_c that corrupts at most $t - 1$ signers of *Thimza*, interacts with the honest signers through the oracles of Definition 4.1 and outputs a pair (msg^*, σ^*) , the probability that $\Sigma'.\text{Verify}(\text{vk}, \text{msg}^*, \sigma^*) = 1$ while msg^* was never submitted to a signing oracle is at most the bound of Theorem 5.5. In particular *Thimza* is existentially unforgeable under chosen message attack from the point of view of a client that runs only $\Sigma'.\text{Verify}$.*

PROOF. By Theorem 5.1 the verification algorithm of *Thimza* is character for character $\Sigma'.\text{Verify}$, so the event in the statement is exactly the event in which the experiment of Definition 4.1 returns 1. The claim is therefore Theorem 5.5 restated, and the bound carries over unchanged. $\square$

The following statement addresses the degenerate case of a single signer.

COROLLARY U.3. *Instantiated with $t = n = 1$, *Thimza* is the Raccoon signature scheme with the commitment truncated by v_c bits, and it is existentially unforgeable under chosen message attack in the quantum random oracle model under MLWE and SelfTargetMSIS.*

PROOF. With $t = n = 1$ the sharing of Algorithm 1 is the identity, so $\mathbf{s}_1 = \mathbf{s}$ and $\lambda_{T,1} = 1$; there is no unordered pair, so the mask of Algorithm 2 is the empty sum and equals zero; and the signer set is the singleton, so the first two rounds carry no information that the signer does not already hold. What remains is $\mathbf{z} = c\mathbf{s} + \mathbf{r}$ together with the hint computed from $\lfloor \mathbf{W} \rfloor_{v_w}$, which is Raccoon signing with $\mathbf{w}$ replaced by $2^{v_c} \lfloor \mathbf{w} \rfloor_{v_c}$. Unforgeability is Theorem 5.5 with an empty corruption set, in which case the view partition of Lemma 4.2 is trivial and the reduction reduces to the one of [13]. $\square$

Remark U.4. Our game does not by itself give TS-UF-1, in which the adversary is allowed to have queried the forgery message as long as it obtained fewer than $t - |C|$ honest partial signatures on it, because the oracle of Figure 1 records a message as signed on the first partial signature. Upgrading to that notion requires counting partial signatures per message inside the reduction and is orthogonal to the two contributions of this paper; the same remark applies to [4, 6], which are also stated at the level of TS-UF-0.